# Geographic coverage debt of frozen Earth-observation foundation models: an encoder-robustness reliability diagnostic and a conditional target-recalibration remedy

Zeyu Fu

***Abstract*—Frozen geospatial foundation models (GFMs) are increasingly deployed as off-the-shelf encoders for land-cover and local-climate-zone (LCZ) monitoring, yet practitioners lack guidance on which frozen encoder to trust after a regional move, or on how to restore a reliability guarantee once shift breaks it. We supply both through a distribution-free reliability audit (split-conformal, finite-sample guarantees) of frozen GFMs under real cross-region shift (thirteen encoders on GEO-Bench BigEarthNet-S2 for the FNR analysis; five on EuroSAT-MS and non-European m-so2sat; the EuroSAT shift is partly a within-Europe label-prior shift, ruled out by a class-balanced control as the sole driver). This audit is packaged as a reusable pre-deployment reliability report-card protocol: for any frozen GFM, it returns four practitioner-facing numbers before deployment—the in-distribution FNR guarantee, the geographic debt that guarantee incurs under shift, the labeled-data cost of repairing that debt with a target-region slice, and the worst-class coverage a marginal guarantee alone would hide. Every frozen encoder meets a multi-label false-negative-rate (FNR) risk-control guarantee in-distribution, yet all thirteen incur a real geographic debt under shift (FNR $2.6$–$4.7\times$ nominal), bought back with larger sets, not a better encoder: FNR robustness rises with set size (Spearman $\rho = -0.66$, $p$=$0.017$), the paper's one adequately powered cross-encoder result. A deterministic boundary bounds when the audit is worth running at all: marginal single-label sets collapse to singletons above accuracy $\approx 0.80$, where the conformal layer adds nothing beyond accuracy. The textbook unlabeled fix does not close the debt—covariate-shift-weighted conformal recovers only part of it, its effective sample size collapsing to $0.1$–$1.6\%$ of the calibration set—so only a labeled target-region slice (e.g. $30\%$ of target data) reliably restores the guarantee, at the cost of large but still-informative restored sets ($22$–$24$ of $\sim$$27$ active labels); this repair is expected by construction, a verification rather than an independent discovery. The cross-encoder question is weaker: the least accurate encoder (Prithvi) is the most FNR-robust, but the rank correlation with accuracy is only a weak, non-significant positive at $n$=$13$ ($\rho = +0.32$, $p$=$0.28$)—directional context, not a proven inversion, since a five-encoder pilot's stronger correlation regressed toward zero on expansion. As a single-encoder, single-class existence proof, restoring the *marginal* coverage guarantee can also hide a per-class collapse invisible to accuracy and ECE. The audit is diagnostic, reusing existing conformal estimators, with preregistration outcome and exploratory-analysis disclosures stated in the main text.**

***Index Terms*—Earth observation, geospatial foundation models, conformal prediction, distribution shift, uncertainty quantification, reliability, calibration.**

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## I. Introduction

GEOSPATIAL foundation models are pretrained once and reused frozen as feature extractors for many downstream tasks. Beyond the encoders audited here (thirteen on BigEarthNet, five on EuroSAT and m-so2sat), frozen GFM backbones now span single-sensor optical pretraining (SatMAE (1), Scale-MAE (2), SeCo (3), GFM (4)), multi-sensor and multi-modal pretraining (SkySense (5), AnySat (6), TerraFM (7), Galileo (8)), and pixel-timeseries pretraining (Presto (9)); none of these families is conformally audited under geographic shift, and our roster (Methods) is illustrative of the diagnostic rather than a leaderboard survey of this rapidly growing space. Their accuracy under geographic shift has been studied (10, 11). Calibration itself is not entirely unstudied: SHRUG-FM (10) evaluates SSL4EO ViT-S/16 (MoCo/MAE/DINO) checkpoints on burn-scar segmentation (ExEBench) and reports expected calibration error (ECE $0.0248$–$0.0328$), but it has no conformal-prediction arm, no spatial-holdout cross-validation, and does not evaluate Clay or Prithvi. Uncertainty quantification for deep networks more broadly spans Bayesian, ensemble, and distribution-free approaches (12); conformal prediction is the distribution-free branch we build on, and it itself has entered Earth observation: Singh et al. (13) bring split and (class-conditional) Mondrian conformal prediction to EO pipelines (Google Earth Engine), GeoConformal (14) adds geographically-weighted conformal intervals for *spatial regression*, and Mao et al. (15) develop model-free conformal prediction for spatially indexed data under local exchangeability; none of the three audits frozen foundation-model encoders, nor coverage loss under an explicit geographic train/deploy shift. Weighted conformal methods for covariate shift (16) are the natural theoretical remedy but require density-ratio estimates we do not assume here. Closest in spirit, Fillioux et al. (17) ask whether frozen vision foundation models are good conformal predictors, but on in-distribution image classification with no geographic-shift audit; and Aolaritei et al. (18) develop Lévy–Prokhorov theory for conformal robustness to distribution shift—complementary theoretical machinery that they do not instantiate on Earth observation. Our geographic-shift framing is also distinct from the broader unsupervised-domain-adaptation literature for remote sensing classification

(19, 20), which typically retrains or adapts the feature extractor itself against the target domain; we instead ask whether a *frozen*, never-retrained encoder's conformal guarantee survives a regional shift, and how to repair it with only a labeled target-region calibration slice, not further training. Consequently, *conformal coverage* reliability under shift—whether a conformal predictor's guarantee (21, 22) on frozen classification GFMs such as Clay and Prithvi survives a move to a new region—has received far less attention. We ask a sharp, falsifiable question: under a real geographic shift, does a conformal audit of a frozen GFM tell a practitioner anything that shifted accuracy and calibration do not? We find that it does, in two regimes, and we characterize precisely where it does not. Our primary contribution is a reusable pre-deployment reliability report-card protocol (§IV-C, Table XI): apply one fixed audit to any frozen GFM and read off four practitioner-facing numbers before deployment—its in-distribution FNR guarantee, the geographic debt that guarantee incurs under shift, the labeled-data cost of repairing that debt with a target-region slice, and the worst-class coverage a marginal guarantee alone would hide. We introduce no new conformal estimator; the audited findings below are what make each report-card entry non-obvious and worth running the protocol for. The protocol's first evidence axis is a within-encoder reliability diagnostic that needs no cross-encoder statistics: under a conformal-risk-control (23) bound on the example-averaged false-negative rate (FNR) on GEO-Bench BigEarthNet-S2, *every one* of thirteen frozen encoders meets the FNR guarantee in-distribution yet incurs a real geographic debt under shift (0.130–0.234, 2.6–4.7× nominal), which a labeled target-region slice repairs (0.002–0.036)—a repair that is expected by construction, since the target-calibrated guarantee is a verification, not an independent discovery. This debt is bought back with larger prediction sets, and that trade is the one adequately powered cross-encoder result in the paper: FNR robustness rises with prediction-set size (Spearman $\rho = -0.66$ at $\alpha$=0.10, $p$=0.017; Table I), an operating-point choice no accuracy scalar can name. The protocol's second evidence axis is the singleton-degeneracy boundary, a deterministic property rather than a correlation: the marginal single-label set collapses to a singleton above accuracy $\approx 0.80$, so EuroSAT at full data is an explicit *negative control* bounding the audit's validity regime (So2Sat/BigEarthNet sit below it)—telling a practitioner exactly when a marginal audit of a frozen GFM is worth running at all. The protocol's third evidence axis is a remedy-failure result, and it is what fixes the report card's target-slice repair cost as necessary rather than optional: the textbook unlabeled fix for covariate shift does not close the debt. Covariate-shift-weighted conformal (16) recovers only part of the coverage loss because its weights' effective sample size collapses to 0.1–1.6% of the calibration set—the density ratios are too heavy-tailed to be usable—so only a small labeled target-region slice reliably restores the guarantee; oracle and deployable label-shift reweighting fail outright (worsening coverage for three of four encoders, ruling out a pure class-prior effect and pointing instead to a within-class conditional shift); and spatial-Mondrian conditioning adds nothing measurable over non-spatial target calibration, so the operative axis the protocol reports is access to labeled target-region data, not a new estimator (scope and estimator provenance are stated in §IV-D). Only after these three do we report the weaker cross-encoder accuracy question, as directional context rather than a claim: accuracy is a weak and unreliable guide to *which* encoder is most robust—the least accurate encoder (Prithvi, mAP 0.432) is the most FNR-robust (0.130) and the most accurate is not—but the rank correlation over the full roster is only a weak, non-significant positive (Spearman $\rho = +0.32$, $p$=0.28; Table I)—the strong +0.90 inversion visible in our first five encoders regresses toward zero when the roster is expanded, so we report it as directional evidence, not a proven inversion (§III). As the protocol's fourth evidence axis—a conditional and single-encoder finding that is exactly what the report card's hidden-worst-class-coverage entry is designed to catch—marginal coverage repair can hide a per-class collapse: on EuroSAT, after target-side recalibration restores the *marginal* guarantee, one land-cover class is still covered at only 0.850 under a 0.956 marginal—a failure invisible to accuracy and to ECE, reported as a single load-bearing existence proof, not a population result (Table II). The protocol's fifth, secondary and descriptive evidence axis is an encoder coverage-debt ranking: the magnitude of geographic coverage debt ranks the frozen GFMs (Prithvi > DOFA (24) > SSL4EO-DINO > SSL4EO-MAE > Clay)—though, because the source-split predictor is in the singleton regime, this ordering coincides with the shifted-accuracy ordering and is reported as descriptive, not as evidence the conformal layer beats accuracy. We establish these on a real EuroSAT-MS (25) cross-region split with five frozen GFMs (the original four plus DOFA (24), added on the EuroSAT grid), corroborate them on GEO-Bench BigEarthNet-S2 (26) (a documented dominant-label proxy, a rigorous per-label multi-label conformal analysis, and a conformal-risk-control arm), and—to test generalization beyond Europe—on the non-European, multi-continent GEO-Bench m-so2sat benchmark. Our analysis is preregistered; where the preregistered *universal*-restoration criterion was not met we preserve that outcome verbatim (§IV-D) and report the conditional, encoder-dependent result as its honest reframe. Fig. 1 summarizes the audit pipeline end to end.

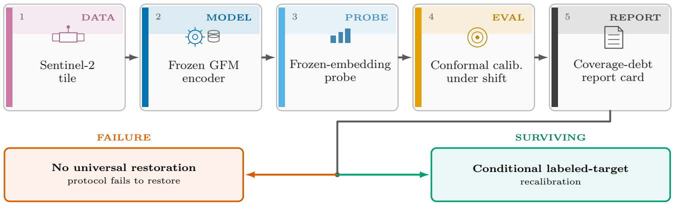


Fig. 1. Overview of the pre-deployment reliability report-card protocol. A Sentinel-2 tile (EuroSAT-MS / GEO-Bench BigEarthNet-S2 / m-so2sat) is passed through one of the frozen, interchangeable GFM encoders under audit (five illustrated: Prithvi, Clay, SSL4EO-DINO, SSL4EO-MAE, DOFA) and a frozen-embedding probe (logistic for the single-label tasks, multi-label otherwise), then conformally calibrated under the real geographic (region-transfer) shift—source split-conformal vs. spatial-Mondrian target-region calibration, source = E/Central Europe, target = W/Iberian-Atlantic Europe—yielding per-class conformal sets. The protocol reads off four practitioner-facing numbers for any frozen GFM: the in-distribution FNR guarantee, the geographic debt under shift, the target-slice repair cost, and the hidden worst-class coverage a marginal guarantee alone would miss. Vermillion marks the failure path (no universal restoration); bluish green marks the surviving path (conditional, encoder-dependent labeled-target recalibration).

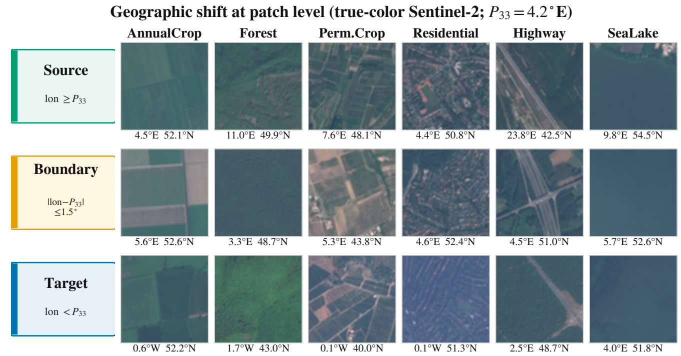


Fig. 3. Geographic shift at patch level: illustrative true-color Sentinel-2 tiles (seeded selection; tile centroids annotated under each patch) for six land-cover classes (AnnualCrop, Forest, PermanentCrop, Residential, Highway, SeaLake) under the frozen $P_{33}$ longitude cut ($= 4.2°$E; source $n$=18,090; target $n$=8,910). Rows encode the split sides: source (lon $\geq P_{33}$), a near-cut boundary band ($|\text{lon} - P_{33}| \leq 1.5°$), and target (lon $< P_{33}$). Region-specific crop structure, settlement texture, and water color differ systematically across the cut — the shift is real but subtle at single-patch level, which is exactly why an accuracy-only deployment check can miss it.

## II. Methods

The data come from EuroSAT-MS (25) (Sentinel-2 L2A, 13 bands, 27,000 tiles, 10 classes). The shift is a *real* geographic cross-region holdout using per-tile longitude: source = E/Central Europe (lon $\geq P_{33}$), target = W/Iberian-Atlantic Europe (lon $< P_{33}$); no image transform is applied (Fig. 2; patch-level examples in Fig. 3). The $P_{33}$ cut is a single, fixed longitude-percentile boundary chosen once in advance; a post-hoc boundary-sensitivity sweep over $P_{25}/P_{40}/P_{50}$ is reported in Results (Table V; see Limitations).

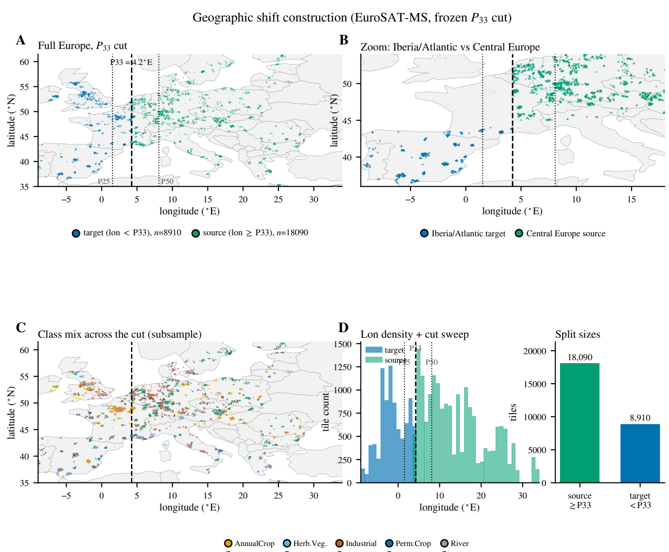


Fig. 2. The geographic-shift construction, shown on the map it is defined on: EuroSAT-MS tile centroids (4,000 of 27,000, seeded subsample) colored by split side at the $P_{33}$ longitude cut ($= 4.2°$E; $P_{25}/P_{50}$ sweep boundaries dotted). Source tiles (E/Central Europe, $n$=18,090) calibrate; target tiles (W/Iberian-Atlantic Europe, $n$=8,910) are the deployment region. No image transform is applied — the shift is purely geographic.

This cut also induces a class-frequency (label-prior) shift between source and target that we did not adjust for in the headline logistic probe: at $P_{33}$, class 5 goes from 2.62% of source tiles to 17.13% of target tiles (6.54× enriched), class 1 from 15.02% to 3.16% (∼ 4.7× depleted), class 8 from 11.60% to 4.51%, and class 9 from 14.10% to 5.04% (remaining classes shift by <2.4×; Table IX, Fig. 14A–B). We re-fit the probe with class-balanced sample weighting on the same cached embeddings as a control (Table X, Fig. 14C–D; see Discussion). The frozen encoders are Prithvi-EO-2.0-300M (27) (mean-pooled patch tokens, 1024-d); Clay v1.5 (28) (cls token, 1024-d); SSL4EO-S12 (29) ViT-S/16 DINO via torchgeo (cls token, 384-d); SSL4EO-S12 (29) ViT-B/16 MAE (`wangyi111/SSL4EO-S12` on HuggingFace, 87.8M params, token-free global pool, 768-d). A fifth encoder, DOFA (Dynamic One-For-All) (24)—a ViT-B/16 whose dynamic wavelength-conditioned patch embedding ingests the Sentinel-2 bands natively (`dofa_base_patch16_224`, `DOFA_MAE` weights via `torchgeo`)—was added on the EuroSAT grid only, frozen and probed identically. The original four weights loaded strictly (zero missing keys), verified real (in-distribution probe accuracy 0.95–0.99 vs. chance 0.10). For the BigEarthNet FNR analysis—the one cross-encoder comparison that turns on statistical power—we expand this battery to *thirteen* frozen encoders by adding eight more loaded through first-party trusted paths (`torchgeo` 0.9.0 built-in weight builders, no third-party repository code executed), each with normalization read from the weight's own shipped transform: SSL4EO-S12 MoCo (ViT-S/16, 384-d) and its ResNet50 MoCo/DINO variants (2048-d) (29); SoftCon (30) ViT-B/14 (768-d) and ViT-S/14 (384-d); an SSL4EO MAE ViT-L/16 (1024-d); and CROMA (31) Base (768-d) and Large (1024-d). Two further planned arms (`dofa_large`, `prithvi_600m`) were dropped by the pre-specified smoke/mAP gates (missing or non-finite feature caches at extraction), leaving thirteen scored; the three anchor encoders reused the frozen caches and reproduce the earlier five-encoder records exactly.

The expanded roster is used only for the BigEarthNet CRC comparison; EuroSAT and m-so2sat retain the five-encoder battery. A logistic probe is trained on frozen embeddings. We compare source-fit split-conformal (21, 22) (LAC (32) nonconformity $1-\hat{p}_{\text{true}}$) against a *spatial-Mondrian* conformal predictor (33) that draws per-spatial-bin quantiles from a disjoint same-region target-calibration slice. We adopt the LAC score for its simplicity; adaptive prediction sets (34) are a natural alternative nonconformity score we do not explore. The Mondrian and target-calibration arms rely on exchangeability within each spatial/target bin, a weaker requirement than global exchangeability; more general non-exchangeable-shift theory (35) formalizes how conformal coverage can degrade under shifts broader than the fixed geographic partition studied here, and the conformal-risk-control arm we use below (23) is itself an instance of the general distribution-free risk-controlling framework of Bates et al. (36). We sweep $\alpha \in \{0.05, 0.10, 0.20\}$, 5 seeds, and bootstrap the paired per-seed restoration statistic $|\text{cov}_{\text{split}} - \text{nom}| - |\text{cov}_{\text{Mondrian}} - \text{nom}|$ (CI $> 0 \Rightarrow$ restored). The set-size cost of that restoration—and the EuroSAT singleton floor it rises above—is reported in Fig. 11. A post-hoc boundary-sensitivity sweep repeats the full protocol at $P_{25}/P_{33}/P_{40}/P_{50}$ longitude cuts on the same cached embeddings, and adds a third, *label-access ablation* arm: non-spatial split conformal whose single LAC quantile is drawn from the same labeled target-region calibration slice the Mondrian arm uses (isolating target-label access from spatial conditioning). All figures (Fig. 9–11, 12, 13, 2, 3, 7, 5, 6, 8, 14) are generated directly from the frozen result records by the accompanying figure-generation scripts. As a post-hoc, exploratory extension, after the preregistered gate (§IV-D) we added three decoupling analyses on the same frozen-feature caches, computed from cached embeddings with no GPU re-extraction: (i) on BigEarthNet-S2, a per-encoder comparison of the CRC arm's shifted FNR against multi-label accuracy (macro mAP) and calibration (mean per-label ECE) analogs on the same footing (the recomputed shifted FNR reproduces the frozen CRC JSON, e.g. Prithvi 0.129 vs. 0.130); (ii) on EuroSAT, per-class conditional coverage of the spatial-Mondrian-repaired target sets at matched marginal coverage ($\approx$ 0.95–0.96), summarized by the worst-class gap $\big((1-\alpha) - \min_k \text{cov}_k\big)$; and (iii) a low-shot sweep that subsamples the EuroSAT probe-training fraction to $\{0.01, \ldots, 1.0\}$ and records split set size and the coverage$-$accuracy gap, mapping the singleton-degeneracy boundary. Correlations across encoders are Spearman rank statistics ($n$=13 FNR on the expanded BigEarthNet roster, $n$=5 conditional coverage) and are reported as directional evidence, not proofs of independence (§IV-D).

## III. Results

In plain terms: every frozen encoder loses reliability under a real geographic move, that loss is repaired only with labeled target-region data, and the sections below map exactly where a conformal audit adds information beyond accuracy and where it does not. A conformal reliability audit is worth its complexity only if it tells a practitioner something accuracy and calibration do not, so we lead with the regimes in which it does and with the one cross-encoder result that is adequately powered. Primary result (next): under a multi-label false-negative-rate (FNR) risk-control guarantee, all thirteen frozen encoders incur a real geographic FNR debt that only a labeled target slice repairs, and that debt is bought back with larger prediction sets (the powered FNR-versus-set-size tradeoff); accuracy itself is only a weak, directional guide to which encoder is most robust. We then map the audit's validity boundary, a deterministic property: in the marginal single-label singleton regime the set-size axis is a restatement of accuracy, so EuroSAT at full data is a mapped negative control, not a decoupling. Next, a remedy-failure result: the textbook unlabeled covariate-shift fix does not close the debt (its weights' effective sample size collapses), so only a labeled target-region slice reliably restores the guarantee. As a conditional, single-encoder finding: on EuroSAT, restoring the *marginal* coverage guarantee hides a per-class conditional-coverage collapse that neither accuracy nor ECE predicts. Only then do we report the encoder coverage-debt ranking and its target-side recalibration repair, as a secondary, descriptive diagnostic. Every axis-decoupling analysis below is post-hoc and exploratory (added after the preregistered gate; §IV-D).

### *A. Primary result: a marginal FNR guarantee incurs a geographic debt no encoder escapes, bought back with larger prediction sets—accuracy is only a weak, directional guide to which encoder is most robust*

On GEO-Bench `m-bigearthnet` (BigEarthNet-S2; 7,669 tiles, 39 active labels; real E$\to$W geographic shift), a conformal-risk-control (CRC) predictor (23) bounds the example-averaged FNR—a set-valued guarantee with *no top-1-accuracy analog by construction*. We audit *thirteen* frozen encoders here (the five-encoder battery plus eight added at pull expressly to power the cross-encoder comparison; roster in §II). The within-encoder finding is unambiguous and requires no cross-encoder statistics: for *every one* of the thirteen encoders, source-calibrated CRC meets the target FNR in-distribution (0.051–0.056 at $\alpha$=0.05) yet incurs a real FNR debt under shift (0.130 Prithvi to 0.234 ResNet50-MoCo, i.e. 2.6–4.7$\times$ nominal across the whole roster), which a target-side Mondrian-CRC arm repays (0.002–0.036; Table I, full per-$\alpha$ CRC/repair numbers in Table VI). Unlike the inert EuroSAT-singleton regime (validity-boundary result below), the repair is active here. This debt-and-repair is the reliability-relevant result: the marginal guarantee a practitioner would trust degrades under a regional move—uniformly, across every backbone we tested—and is restored only with a labeled target-region slice. This pattern is not merely descriptive: treating the thirteen preregistered per-encoder point estimates as a population, a distribution-free sign test confirms that shifted CRC FNR exceeds the nominal target in all thirteen encoders (two-sided $p \approx 2.4 \times 10^{-4}$) and that target-side Mondrian-CRC returns FNR to at or below nominal in all thirteen (same test, same $p$); a Wilcoxon signed-rank sensitivity check agrees exactly (one-sided $p \approx 1.2 \times 10^{-4}$ for both directions, the smallest value attainable at $n$=13), and every one of the thirteen per-encoder bootstrap confidence intervals excludes nominal in the claimed direction, for both the debt and the repair arm. We report the one-sided figure only as a sensitivity check: the debt-and-repair direction was not discovered in this roster run but inherited from the original five-encoder confirmatory arm, so it is defensible, but the two-sided value is what we use as the headline since it needs no such defense and is already decisive. This is a population-level summary computed post-hoc over the frozen, preregistered per-encoder estimates—the per-encoder numbers are prereg and frozen, but the population test itself is not—and it supplies a second firm, statistically-tested positive result alongside the preregistered falsification of universal restoration (§IV-D). Two further qualifications bound what the test shows. First, the thirteen encoders share the same target-shift BigEarthNet split and labels, so this is a distribution-free summary over thirteen preregistered per-encoder point estimates, not an i.i.d.-sample hypothesis test: the nominal $p$ treats the sign pattern combinatorially, and shared-target dependence means the effective number of independent draws is smaller than thirteen. Second, the repair-arm direction is what the target-calibrated Mondrian-CRC guarantee is constructed to deliver, so its $13/13$ reads as a verification that the guarantee holds in practice rather than as an independent discovery. Re-aggregating to the level of independent encoder families—encoders sharing a pretraining corpus and method lineage (all six SSL4EO-S12-pretrained ViT/ResNet variants as one family; SoftCon and CROMA each one family across their two size variants; Prithvi, Clay, and DOFA as singleton families, six families in total) and requiring unanimous within-family agreement, the most conservative vote rule available—still gives a unanimous $6/6$ sign test in both directions, but at a far weaker two-sided $p \approx 0.031$ (one-sided $p \approx 0.016$): the family-clustered correction survives at $\alpha$=0.05, but only barely, so the thirteen per-encoder bootstrap confidence intervals (all excluding nominal in the claimed direction) remain the primary evidence, not the population $p$-value at either $n$.

The one adequately powered cross-encoder result is a risk-versus-efficiency tradeoff: FNR robustness is bought with larger prediction sets ($\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \text{set size}) = -0.33$ at $\alpha$=0.05, $-0.66$ ($p$=0.017) at $\alpha$=0.10; Prithvi buys the best FNR with the largest sets, 15.5 labels/tile)—an operating-point choice a practitioner can act on directly, independent of which encoder they picked.

Does accuracy also tell a practitioner *which* encoder to trust? Here the evidence is weaker, and we report it as directional context rather than a claim. The FNR under shift is only weakly rank-correlated with in-distribution mAP ($\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \mathrm{mAP}) = +0.32$, permutation $p$=0.28, $n$=13, $\alpha$=0.05; Fig. 4) and not at all with calibration ($\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \mathrm{ECE}_{\mathrm{in}}) = -0.16$, $p$=0.60). The correlation is *positive*—higher accuracy never buys better FNR robustness, and the ordering never reverses to “better encoder $\rightarrow$ better reliability”—but it does not reach significance and 48 of 78 encoder pairs (62%) invert, so we do *not* claim a strong accuracy-to-reliability inversion; at $n$=13 this is a weak, directional statement, not proof that accuracy is uninformative. One extreme survives the expansion and carries the practical point: Prithvi, the least accurate encoder (mAP 0.432), is the *most* FNR-robust (0.130), while the two most FNR-fragile encoders—the ResNet50 CNNs (MoCo/DINO, 0.234/0.232)—sit mid-pack on accuracy; and Prithvi vs. Clay is a near-calibration-matched dissociation (ECE 0.082 vs. 0.078, differing by 0.004, yet FNR 0.130 vs. 0.176, the worse-calibrated encoder holding the better FNR). In those first five encoders the same mAP-correlation statistic read +0.90 (asymptotic $p$=0.037; its *exact* permutation $p$ was already 0.083); the roster expansion regresses it toward zero, precisely the power test the expansion was run to perform, and we report the weakened result rather than the headline it displaced (§IV-D).

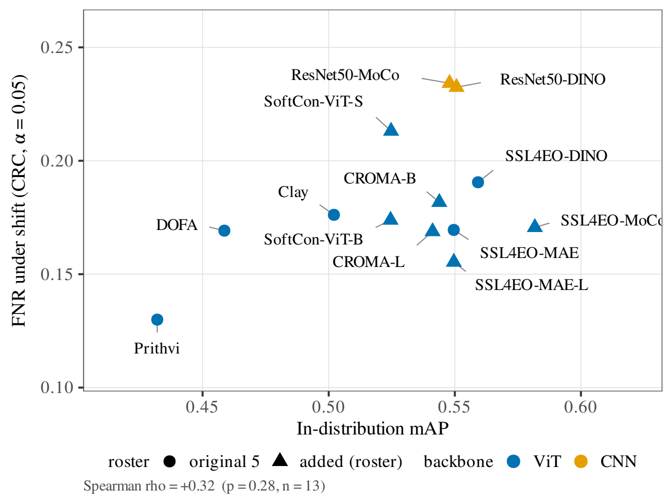


Fig. 4. The five-encoder inversion does not survive expansion to thirteen. FNR under shift vs. in-distribution mAP on `m-bigearthnet` ($\alpha$=0.05, CRC arm); circles are the original five encoders, triangles the eight added at pull. The rank correlation is a weak, non-significant positive (Spearman $= +0.32$, permutation $p$=0.28, $n$=13; no trend line is drawn, since the correlation does not reach significance): accuracy never buys better FNR robustness, but the strong +0.90 inversion of the five-encoder slice regresses toward zero. Prithvi (least accurate) remains the most FNR-robust; the two ResNet50 CNNs are the most fragile at mid accuracy.

TABLE I

PRIMARY DECOUPLING ON GEO-BENCH M-BIGEARTHNET ($\alpha$=0.05; 13 FROZEN ENCODERS, SORTED BY MAP): EVERY ENCODER MEETS THE TARGET FNR IN-DISTRIBUTION AND INCURS A 2.6–4.7× FNR DEBT UNDER SHIFT THAT THE TARGET-SIDE MONDRIAN-CRC ARM REPAIRS. "FNR SHIFT" IS THE SOURCE-CALIBRATED CRC ARM'S SHIFTED FNR; "REPAIRED" IS THE TARGET-SIDE MONDRIAN-CRC ARM; SET SIZE IS CRC LABELS/TILE UNDER SHIFT. ACCURACY IS A WEAK, NON-SIGNIFICANT GUIDE TO FNR ROBUSTNESS: $\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \mathrm{mAP}) = +0.32$ ($p$=0.28; 48/78 INVERSIONS), $(\mathrm{FNR}_{\mathrm{sh}}, \mathrm{ECE}) = -0.16$ ($p$=0.60), $(\mathrm{FNR}_{\mathrm{sh}}, \text{SET SIZE}) = -0.33$ ($n$=13; −0.66, $p$=0.017 AT $\alpha$=0.10).

| Encoder | mAP (in) | ECE (in) | FNR shift | FNR repaired | Set size |
|---|---|---|---|---|---|
| Prithvi | 0.432 | 0.082 | 0.130 | 0.031 | 15.5 |
| DOFA | 0.459 | 0.057 | 0.169 | 0.021 | 13.0 |
| Clay | 0.502 | 0.078 | 0.176 | 0.022 | 13.3 |
| SoftCon-ViT-B | 0.525 | 0.033 | 0.174 | 0.031 | 11.9 |
| SoftCon-ViT-S | 0.525 | 0.020 | 0.213 | 0.036 | 10.7 |
| CROMA-L | 0.541 | 0.037 | 0.169 | 0.024 | 11.6 |
| CROMA-B | 0.544 | 0.046 | 0.182 | 0.030 | 12.1 |
| ResNet50-MoCo | 0.548 | 0.073 | 0.234 | 0.002 | 12.5 |
| SSL4EO-MAE | 0.550 | 0.037 | 0.170 | 0.030 | 11.8 |
| SSL4EO-MAE (ViT-L) | 0.550 | 0.038 | 0.155 | 0.029 | 11.9 |
| ResNet50-DINO | 0.551 | 0.055 | 0.232 | 0.014 | 10.8 |
| SSL4EO-DINO | 0.559 | 0.028 | 0.191 | 0.032 | 10.6 |
| SSL4EO-MoCo | 0.582 | 0.032 | 0.171 | 0.028 | 10.7 |

### *B. Secondary (conditional) result: marginal coverage repair hides a per-class conditional collapse*

After the target-side spatial-Mondrian arm restores the *marginal* EuroSAT guarantee to $\approx$ 0.95–0.96 for every encoder (secondary results below), we ask what that marginal scalar hides: how badly does the worst individual land-cover class still under-cover? At matched marginal coverage ($\alpha$=0.05, 5 seeds; Table II; Fig. 5) the worst-class gap is scrambled relative to accuracy and ECE—$\mathrm{Spearman}(\text{worst gap}, \text{acc}) = +0.30$ ($p$=0.62), $\mathrm{Spearman}(\text{worst gap}, \mathrm{ECE}) = -0.30$ ($p$=0.62), 6 of 10 pairwise inversions against each, partial correlation $\approx 0$ ($-0.03$ for acc$|$ece). The smoking gun: SSL4EO-DINO is mid-pack on accuracy ($0.880$, 3rd of 5) and ECE ($0.076$, 3rd) yet silently abandons a class—one EuroSAT class is covered at only $0.850$ under a $0.956$ marginal (a $0.106$ within-encoder conditional shortfall, $2\times$ the next-worst encoder's gap), and it is the same class that under-covers in all five seeds (per-seed coverage $0.826$–$0.871$, std $0.016$; the datum is seed-stable, not a single-seed artifact). Nothing in its accuracy or calibration predicts this: DOFA, the second-least accurate encoder ($0.851$), has the best conditional coverage (gap $0.031$), while Clay—the most accurate ($0.918$) and best-calibrated (ECE $0.057$)—has the second-worst (gap $0.048$). The effect sharpens at $\alpha$=0.10, where SSL4EO-DINO's worst class sits at $0.776$ under a $0.926$ marginal. Which frozen encoder quietly drops a class under geographic shift is invisible to accuracy and to ECE and is surfaced only by the conditional conformal audit. Honest caveat: $n$=5 gives low power—all $p > 0.28$ and the weak rank correlation even flips sign across $\alpha$—so the claim is "no detectable/stable dependence plus large concrete inversions," not proven independence; the SSL4EO-DINO case is the load-bearing single datum and is reported as such (§IV-D).

TABLE II
PER-CLASS CONDITIONAL COVERAGE OF THE MARGINALLY REPAIRED EUROSAT PREDICTOR (SPATIAL-MONDRIAN, $\alpha$=0.05; MARGINAL COVERAGE MATCHED TO $\approx$ 0.95–0.96; 5 SEEDS). THE WORST-CLASS GAP DOES NOT TRACK ACCURACY OR ECE (Spearman = +0.30/ − 0.30, $p$=0.62; 6/10 INVERSIONS). SSL4EO-DINO, MID-PACK ON BOTH SCALARS, ABANDONS A CLASS AT 0.850 (SEED-STABLE; RANGE 0.826–0.871); THE MOST ACCURATE AND BEST-CALIBRATED ENCODER (CLAY) HAS THE SECOND-WORST GAP. ACCURACY/ECE COLUMNS ARE THE CANONICAL SHIFT VALUES; THE CONDITIONAL COLUMNS ARE THE EXP0 RECOMPUTE.

| Encoder | Acc. shift | ECE shift | Marg. cov | Worst-class cov | Worst gap (vs. nominal) |
|---|---|---|---|---|---|
| DOFA | 0.851 | 0.093 | 0.951 | 0.919 | 0.031 |
| SSL4EO-MAE | 0.898 | 0.065 | 0.962 | 0.912 | 0.038 |
| Prithvi | 0.833 | 0.109 | 0.955 | 0.905 | 0.045 |
| Clay | 0.918 | 0.057 | 0.960 | 0.902 | 0.048 |
| SSL4EO-DINO | 0.880 | 0.076 | 0.956 | 0.850 | 0.100 |

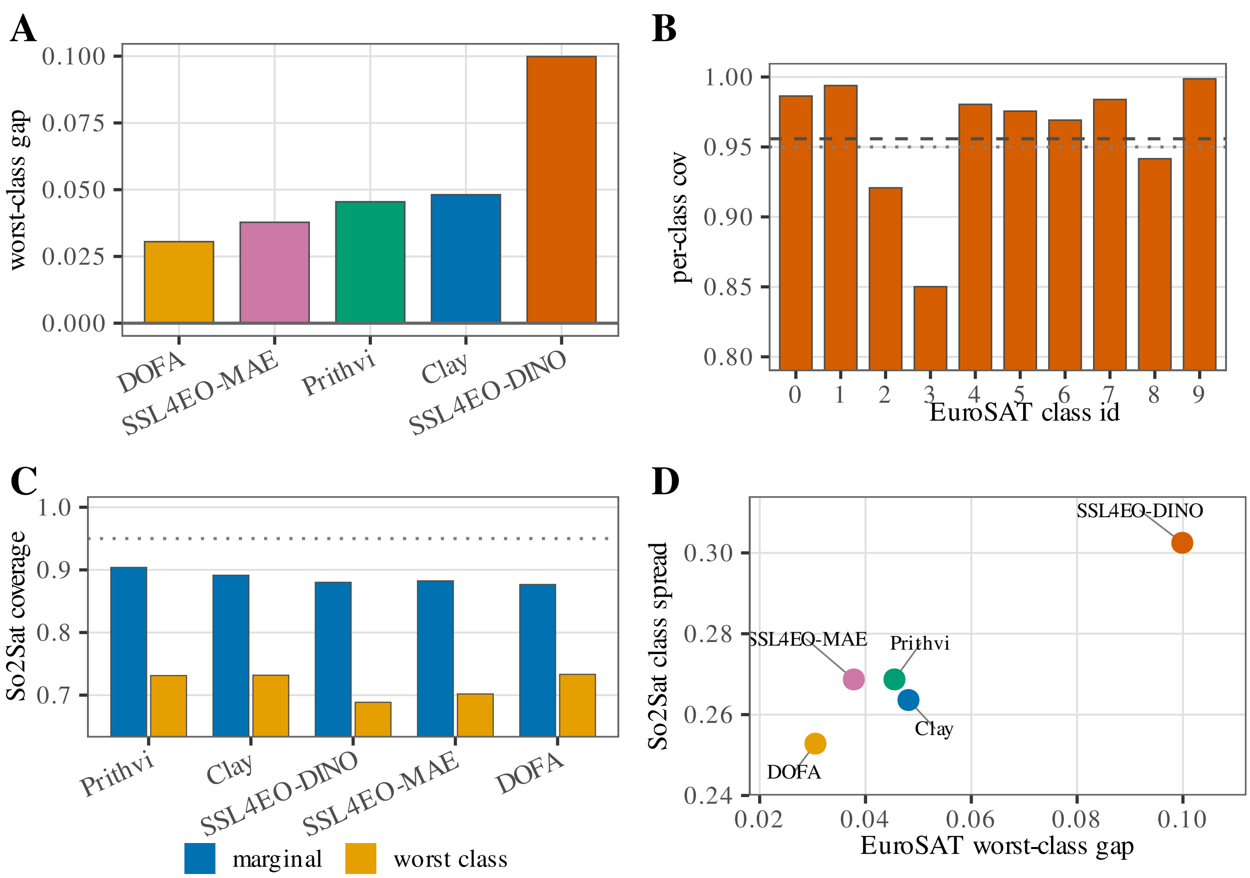


Fig. 5. Conditional-coverage panels from the frozen EuroSAT and So2Sat roster summaries (no new runs; $\alpha$=0.05). (A) EuroSAT worst-class coverage gap $G_{\text{worst}}$ by encoder (Okabe–Ito colours). (B) SSL4EO-DINO per-class conditional coverage on EuroSAT (dashed = encoder marginal; dotted = nominal 0.95). (C) So2Sat source-only split coverage by encoder: marginal (blue) vs. worst class (gold); dotted = nominal 0.95. (D) Encoder scatter of EuroSAT worst-class gap vs. So2Sat class-coverage spread (same encoder colours as (A)).

### *C. The audit's validity boundary: singleton degeneracy of the marginal single-label set*

A conformal set-size axis can decouple from accuracy only where the sets are non-trivial. On EuroSAT at full data the source-calibrated split predictor emits *singletons* on the shifted target (mean set size 1.00 for every encoder and $\alpha$; per-seed $\leq$ 1.001; Fig. 11b), so its coverage coincides with shifted top-1 accuracy (5-seed-mean $|\Delta| \leq 2 \times 10^{-4}$) and the marginal set-size/efficiency axis carries no information beyond accuracy. Even after target-side repair the restored Mondrian set size is a deterministic image of accuracy loss (Fig. 11a): across the five encoders $\mathrm{Spearman}(\text{set size}, \text{acc}) = -1.0$ and $\mathrm{Spearman}(\text{set size}, \text{ECE}) = +1.0$ ($\alpha$=0.05, zero pairwise inversions). We therefore report set size only as a companion to a held guarantee, never as evidence that the conformal layer beats accuracy—the preregistered "efficiency-decoupling" route is empirically falsified on this benchmark and we say so. Mapping the boundary. Subsampling the probe-training fraction drives shifted accuracy down and locates where the audit switches from degenerate to informative

(Table III, Fig. 6). In the high-accuracy regime (acc $\gtrsim 0.80$ at $\alpha$=0.05) sets are exact singletons (size $\equiv 1.000$) and coverage $\equiv$ accuracy; as the task hardens below acc $\approx 0.78$ the sets go non-singleton (size $> 1.05$) and the coverage$-$accuracy gap opens from $0$ to $+0.22$. The more accurate the encoder, the higher the accuracy at which it degenerates (Clay singleton by acc $\approx$ $0.87$, SSL4EO-DINO by $\approx$ $0.82$). This makes EuroSAT at full data an explicit negative control: it sits above the boundary, so its *marginal* single-label audit is accuracy-restated—which is precisely why the per-class conditional audit above (on the non-singleton repaired sets) and the multi-label FNR audit on BigEarthNet are where the conformal layer earns its keep, and why So2Sat (17-class LCZ, shifted accuracy $0.52$–$0.60$), which sits below the boundary, carries informative non-singleton sets even marginally (Table VII). The boundary is itself a contribution: it tells a practitioner when a marginal conformal audit of a frozen GFM is worth running.

TABLE III

SINGLETON-DEGENERACY BOUNDARY (EUROSAT, PRITHVI, $\alpha$=0.05, 3 SEEDS): SUBSAMPLING THE PROBE-TRAINING FRACTION LOWERS SHIFTED ACCURACY AND MOVES THE SPLIT PREDICTOR FROM DEGENERATE (SINGLETONS, COVERAGE $\equiv$ ACCURACY) TO INFORMATIVE (NON-SINGLETON SETS, COVERAGE DECOUPLED FROM ACCURACY). THE TRANSITION SITS NEAR ACCURACY $\approx$ 0.78–0.80.

| Train frac | Train $n$ | Shifted acc | Split set size | Coverage $-$ accuracy |
|---|---|---|---|---|
| 1.00 | 10854 | 0.832 | 1.000 | +0.000 |
| 0.50 | 5427 | 0.805 | 1.026 | +0.011 |
| 0.25 | 2713 | 0.783 | 1.072 | +0.026 |
| 0.10 | 1085 | 0.765 | 1.202 | +0.066 |
| 0.05 | 542 | 0.721 | 1.383 | +0.107 |
| 0.02 | 217 | 0.645 | 1.866 | +0.196 |
| 0.01 | 108 | 0.650 | 2.294 | +0.221 |

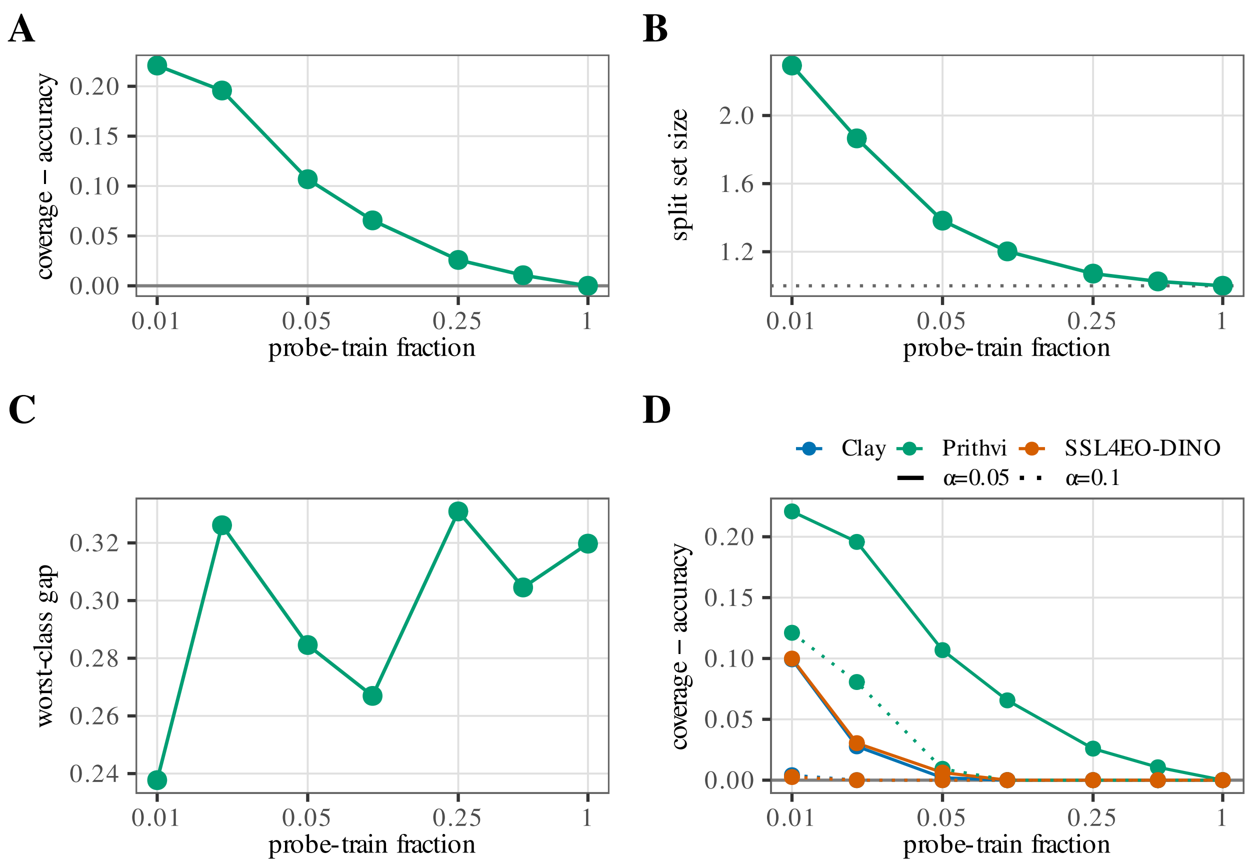


Fig. 6. Singleton / low-shot validity boundary from the frozen EuroSAT low-shot sweep (no new runs). Panels (A)–(C) show the Prithvi series at $\alpha$=0.05 vs. probe-train fraction (log-$x$). (A) Coverage−accuracy gap: as the budget rises, gap$\to$ 0 and coverage collapses onto accuracy (degenerate regime). (B) Mean split set size on the same $x$-axis—a *different* quantity from (A), not a restatement of the gap; the dotted line at size = 1 is the singleton floor. (C) Worst-class coverage gap vs. fraction (conditional residual under the same sweep). (D) Multi-encoder comparison (Clay, Prithvi, SSL4EO-DINO): coverage−accuracy vs. fraction with colour = encoder and linestyle solid $\alpha$=0.05 / dotted $\alpha$=0.10. Panels (A) and (B) are related through the singleton transition but are not redundant: gap measures coverage–accuracy decoupling, while set size measures how large the prediction sets become once that decoupling appears.

*D. Secondary descriptive: the coverage-debt encoder ranking and its target-side repair*

The remaining EuroSAT results are the coverage-debt ranking and target-recalibration repair. They are a useful descriptive diagnostic but, per the validity-boundary result above, the marginal ranking coincides with the shifted-accuracy ranking; we report them as such, not as an independent conformal signal. Coverage debt is real and encoder-dependent: under shift, accuracy drops $0.95 \to 0.83$ (Prithvi), $0.97 \to 0.88$ (SSL4EO-DINO), $0.98 \to 0.92$ (Clay), and ECE rises to 0.109 (Prithvi), 0.076 (SSL4EO-DINO), 0.065 (SSL4EO-MAE), 0.057 (Clay)—an ordering of calibration degradation across the four encoders that is preserved under the boundary sweep (Table V; Fig. 9C) and the class-prior-balanced probe control (Discussion), so it does not rest on a bare four-point monotonicity (Fig. 9A–B). As a continuous, bucket-free debt measure we integrate the target coverage gap $\max(0, (1 - \alpha) - \mathrm{cov}_{\mathrm{split}})$ over the $\alpha$ sweep: $D = 8.0/3.3/1.4/0.8 \times 10^{-3}$

for Prithvi / SSL4EO-DINO / SSL4EO-MAE / Clay—and $D \approx 6.2\times10^{-3}$ for DOFA, added on the same grid as a fifth encoder and the second-heaviest debt, between Prithvi and SSL4EO-DINO—recovering the same ordering without the coarse $k/3$ repaired-levels bucketing. The gap closes at $\alpha = 0.20$ for all five encoders because shifted accuracy already meets the $0.80$ nominal—a no-debt artifact of the easy risk level, not evidence of a repair. DOFA's shifted split coverage is $0.851$ at every risk level, so like the four battery encoders it carries under-coverage debt only at $\alpha = 0.05$ and $0.10$.

Spatial-Mondrian repays the debt where it exists: at $\alpha = 0.05$ (nominal $0.95$), source split-conformal under-covers ($0.833$ Prithvi, $0.851$ DOFA, $0.880$ SSL4EO-DINO, $0.898$ SSL4EO-MAE,[1] $0.918$ Clay) while spatial-Mondrian restores coverage to $0.955$, $0.951$, $0.956$, $0.962$, and $0.960$ respectively, with paired-bootstrap CI excluding $0$ for all five encoders (Fig. 10). The magnitude of repair tracks the debt: at this shared risk level the restoration set size is monotone in debt—$1.62$ (Prithvi) $>$ $1.58$ (DOFA) $>$ $1.43$ (SSL4EO-DINO) $>$ $1.39$ (SSL4EO-MAE) $>$ $1.22$ (Clay) (Fig. 11). The number of risk levels repaired is weakly monotone: Prithvi (heaviest debt) and DOFA (second-heaviest) are each repaired at $2/3$ levels ($\alpha = 0.05, 0.10$), while SSL4EO-DINO, SSL4EO-MAE, and Clay are each repaired at $1/3$ ($\alpha = 0.05$). At $\alpha = 0.20$ (nominal $0.80$) none of the five is repaired because shifted accuracy ($0.83$–$0.92$, DOFA $0.85$) already meets nominal—there is no under-coverage debt to repay, and the correction correctly does nothing (Fig. 10). DOFA was added to the same EuroSAT grid as a fifth encoder in a single-FM run whose spatial-Mondrian set sizes are $1.58/1.29/1.08$ at $\alpha = 0.05/0.10/0.20$; an aggregate fail under the multi-encoder universal-restoration rule is expected here (Clay and Prithvi were absent from that single-encoder call), while the per-encoder outcome is restoration at $\alpha = 0.05$ and $0.10$ by the same paired-bootstrap CI-excludes-0 test applied to the four battery encoders (paired gap-difference CI low $>$ 0), and no repair at $\alpha = 0.20$, where DOFA already over-covers ($0.851$). Correction note: this DOFA arm was re-run after an initial run inadvertently probed DOFA's $45$-way classification-head output instead of its 768-d pooled feature embedding; all DOFA numbers reported here are from the corrected embedding run, with the incorrect probe outputs archived for provenance. The four battery encoders are unaffected: their cached features were verified at their documented embedding widths (Prithvi and Clay 1024-d, SSL4EO-DINO 384-d, SSL4EO-MAE 768-d), and the bug's $45$-dimensional signature appears in no other feature cache.

Temperature scaling (37) proves insufficient: applying the source-fit temperature to the shifted logits lowers ECE to $0.058/0.041/0.054/0.032$ (Prithvi/SSL4EO-DINO/SSL4EO-MAE/Clay) from the raw $0.109/0.076/0.065/0.057$, but the residual still sits 2–11$\times$ above the in-distribution level ($0.026/0.014/0.005/0.013$), motivating the distribution-free conformal route; a focal-loss-based calibration alternative (38) is likewise post-hoc-parametric and is not explored here.

The headline is not a single-cut artifact: sweeping the source/target longitude cut over $P_{25}/P_{33}/P_{40}/P_{50}$ on the same cached embeddings (5 seeds each; Table V) preserves the shift-ECE encoder ordering Prithvi $>$ SSL4EO-DINO $>$ SSL4EO-MAE $>$ Clay at every boundary (Prithvi $0.144/0.108/0.103/0.093$ vs. Clay $0.074/0.057/0.051/0.041$ at $P_{25}/P_{33}/P_{40}/P_{50}$), and the debt shrinks monotonically as the cut moves east and the target becomes less extreme (Prithvi split coverage $0.792 \to 0.859$). The ordering is over a small roster (five encoders at $P_{33}$, four across the sweep) and some adjacent gaps sit within seed noise—e.g. SSL4EO-DINO and SSL4EO-MAE differ by only $\approx$0.001 shift-ECE at $P_{25}$—so the extremes (Prithvi heaviest, Clay lightest) are robust while adjacent mid-rank ties should not be over-interpreted. At $\alpha = 0.05$, Mondrian restoration (paired-seed CI excluding 0) holds in 15 of 16 encoder$\times$boundary cells; the sole exception is Clay at $P_{50}$, where split coverage is already $0.936$ (debt $\approx$0.014)—exactly the debt-targeted behaviour the conditional framing predicts. At $\alpha = 0.10$, Prithvi (heaviest debt) is restored at all four boundaries while the lighter-debt encoders mostly are not.

A label-access ablation shows that target calibration, not spatial binning, is the active ingredient: the non-spatial target-calibrated arm restores coverage to $0.946$–$0.952$ at $\alpha = 0.05$ in every debt cell—statistically indistinguishable from spatial-Mondrian: the paired-seed "Mondrian minus target-global" gap-reduction CI excludes 0 in zero of 48 (FM$\times$boundary$\times\alpha$) cells, while target-global itself beats source-only split-conformal in the 22 cells where debt exists. Mondrian's set sizes are consistently larger ($P_{33}$, $\alpha = 0.05$: 1.62 vs. 1.56 Prithvi; 1.44 vs. 1.30 SSL4EO-DINO; 1.39 vs. 1.20 SSL4EO-MAE; 1.22 vs. 1.12 Clay). On this benchmark and $4 \times 4$ grid, then, the repair is bought by access to labeled target-region calibration data; per-spatial-bin conditioning adds no measurable marginal coverage benefit. We report this honestly and adjust the claim: the operative correction is *target-side conformal recalibration*, of which spatial-Mondrian is one (slightly costlier) instantiation here; finer spatial heterogeneity, where within-target variation is stronger, may still separate the two arms. Once spatial conditioning is inert, the remedy reduces to standard split-conformal calibrated on a small labeled target-region slice: we introduce no new conformal method, and spatial binning is reported here as an honest negative. The contributions of this paper are correspondingly the encoder-robustness diagnostic and the preregistered audit, not a new estimator.

How small a labeled target-region slice does the repair require under this Cross-European optical shift? The frozen protocol prices the remedy at a single point—the fixed 30% target-calibration fraction—so we sweep the budget post-hoc over $0.5\%$–$30\%$ of the target region on the same cached embeddings, holding the evaluation slice fixed at the frozen cut (Fig. 7, *descriptive, not confirmatory*). The repaired arm clears the nominal level at the smallest swept budget and then *decreases* toward nominal as the budget grows: finite-sample conformal conservatism, not a gradual repair. At $n$=44 the effective quantile $\lceil (n+1)(1-\alpha) \rceil / n$ is $0.932$ vs. nominal

[1]SSL4EO-MAE's shifted split coverage is 0.8984 in this five-encoder run (shown as 0.898) and 0.8985 in the covariate- and label-shift baseline runs (Tables IV–IV, shown as 0.899); the $<10^{-4}$ difference is independent calibration-split RNG on the same cached features, and split coverage equals shifted top-1 accuracy in every case (singleton sets).

$0.900$ at $\alpha$=$0.10$, over-covering and paying for the slack in larger sets; as the budget grows toward $30\%$ the inflation shrinks and coverage converges onto nominal—from far above it at the cheapest budget (Prithvi at $0.5\%$: $0.994/0.959/0.904$ at $\alpha$=$0.05/0.10/0.20$) down to the frozen operating point's $0.952/0.899/0.833$. For a practitioner the labeled budget therefore controls set-size tightness, not coverage validity: wherever debt exists, any labeled slice—down to the finite-sample floor—already repairs the guarantee. The exchange rate is steep: for Prithvi, moving from the frozen $n$=$2673$ down to $n$=$44$—a $61\times$ annotation saving—inflates the mean set from $1.56$ to $3.66$ at $\alpha$=$0.05$ ($+134\%$), from $1.22$ to $1.77$ at $\alpha$=$0.10$ ($+46\%$), and from $1.00$ to $1.28$ at $\alpha$=$0.20$ ($+28\%$). Clay at $\alpha$=$0.20$ is flat at the split value $0.9180$ at every budget (singleton regime; Table III). The practical consequence is a deflationary one for the $30\%$ figure the report card quotes: it is a *sufficient* budget, not a necessary one. The BigEarthNet CRC arm shows the same structure with a hard floor: CRC can only certify risk $\alpha$ when $\alpha \geq 1/(n_{\text{eff}}+1)$, so at $\alpha$=$0.05$ at least 19 positive-label calibration examples are needed; below that the set becomes every label and measured FNR collapses to zero vacuously (excluded rather than reported as a free repair). Above the floor, FNR rises toward the target as the budget grows (e.g. at $\alpha$=$0.05$, $0.022$ at $1\%$ against $0.051$ at $30\%$), while $\approx 1\%$ of the target region already discharges the risk debt and the remaining annotation buys tighter sets.

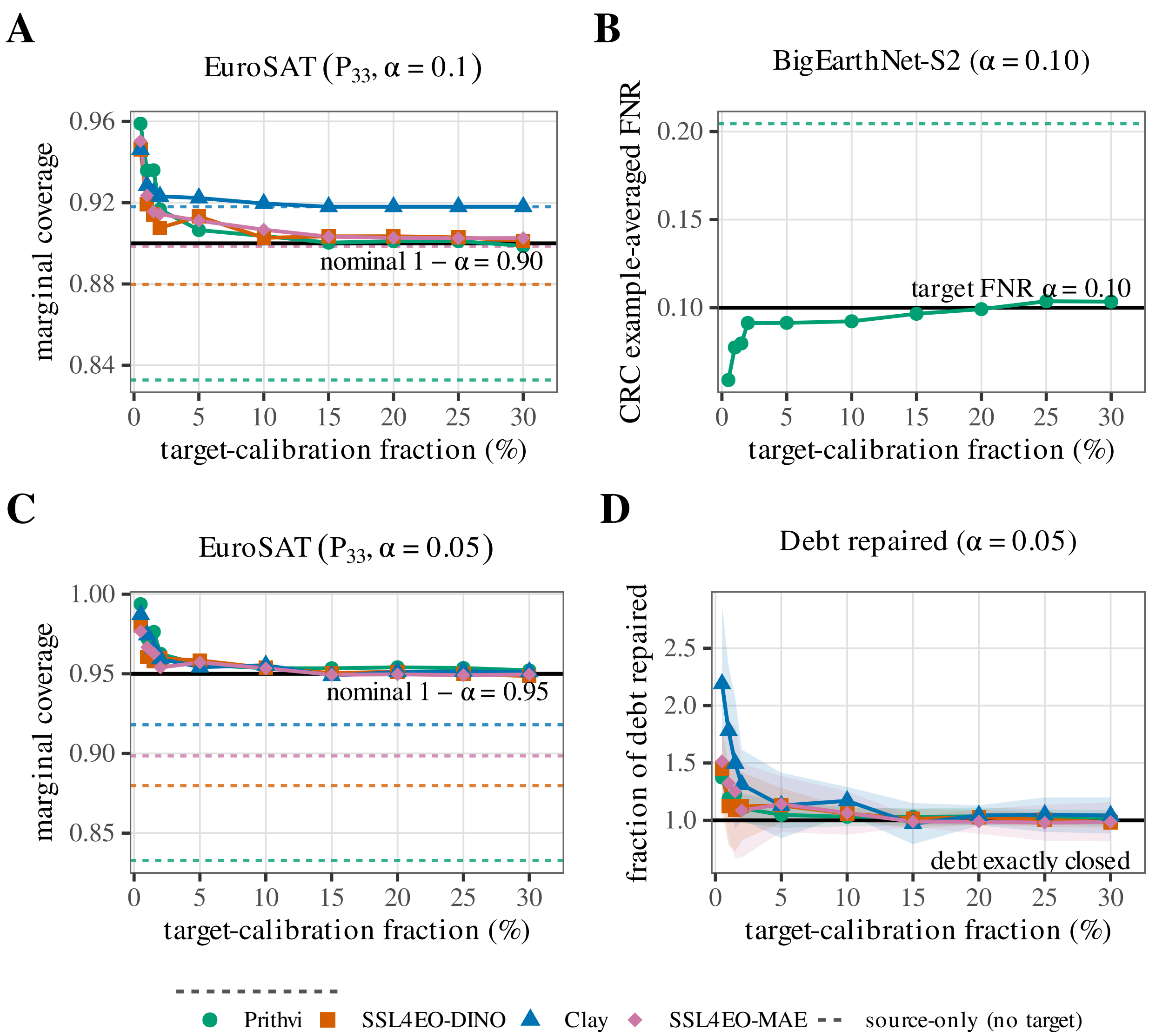


Fig. 7. Target-recalibration planning curves: annotation budget versus guarantee tightness. **(A)** EuroSAT-MS ($P_{33}$, $\alpha$=0.10): marginal coverage of the target-calibrated arm versus the labeled fraction of the target region (%). **(B)** GEO-Bench BigEarthNet-S2 ($\alpha$=0.10): CRC example-averaged false-negative rate (FNR) on the same $x$-axis (Prithvi only; vacuous CRC budgets with $\alpha < 1/(n_{\text{eff}}+1)$ omitted). **(C)** EuroSAT-MS at the stricter $\alpha$=0.05, same coverage axes as (A). **(D)** Normalized EuroSAT reading at $\alpha$=0.05: fraction of coverage debt repaired versus budget (shaded bands = five-seed intervals; values $>1$ are over-repair). Solid coloured curves are encoders; matching dashed horizontals are budget-independent source-only baselines (no target labels); black horizontals mark the nominal coverage $1-\alpha$, target FNR $\alpha$, or exact debt clearance (= 1 in (D)). The repair is not a rising ramp: wherever debt exists, the smallest usable budget already clears the nominal level, after which curves converge onto it from the conservative side. That shape is finite-sample conformal correction—the effective quantile $\lceil (n+1)(1-\alpha) \rceil / n$ inflates above $1-\alpha$ at small $n$ (e.g. 0.932 vs. 0.900 at $n$=44, $\alpha$=0.10), over-covering and paying for the slack in larger sets—so extra labels buy tightness to nominal, not validity of the guarantee. Descriptive secondary analysis; So2Sat excluded (no local features).

Tables IV–VII are a robustness battery, not five independent findings: each holds the frozen encoders and the coverage debt fixed and swaps exactly one variable—the correction method (unlabeled covariate-shift reweighting in Table IV; label-shift-adjusted reweighting in Table IV), the source/target geographic cut (Table V), or the benchmark itself (BigEarthNet's multi-label FNR arm in Table VI; the non-European So2Sat benchmark in Table VII). In every table the single comparison worth making is the same one: at the shared $\alpha = 0.05$ row, read the source-only ("split") figure against the labeled target-side repair figure in that row—the gap between them is the debt each table is stress-testing, and it is the only number that would change the paper's claim if it closed.

A stronger unlabeled baseline still does not close the debt. Is the labeled target slice truly necessary, or would a principled unlabeled remedy suffice? As a post-hoc exploratory baseline on the cached EuroSAT features we add covariate-shift-weighted split conformal (16): density-ratio weights $w(x)$ from a logistic source-vs-target discriminator on the frozen embeddings, weighted-quantile calibration on the source scores, scored on the same shifted target (Table IV). Reweighting recovers part of the debt—at $\alpha = 0.05$ target coverage rises from source-split $0.833$–$0.918$ to $0.911$–$0.938$ per encoder—but under-covers the nominal $0.95$ for every encoder and never matches the labeled target-global arm ($\approx 0.95$). The mechanism is visible in the weights: the effective sample size is only $0.1$–$1.6\%$ of the calibration set — density ratios so heavy-tailed that a handful of points dominate, a direct signature that this real E$\rightarrow$W shift is not the pure covariate shift under which weighted conformal is guaranteed (it also carries the class-prior shift documented in Methods). Even a principled unlabeled correction is thus insufficient and unstable here; a small labeled target slice remains the reliable remedy.

A label-shift-adjusted control rules out the class-prior confound: the $P_{33}$ cut's severe class-prior shift (Table IX) raises the possibility that the coverage debt above is a textbook label-shift effect (39) rather than a genuinely geographic one. We test this directly at the identical $P_{33}$ split: we reweight the source calibration scores by $w(y) = \hat{\pi}_{\text{target}}(y)/\hat{\pi}_{\text{source}}(y)$ (Podkopaev–Ramdas label-shift-weighted split conformal), in an *oracle* variant (the true target-region label marginal, an upper bound on what prior-shift correction alone could repay) and a deployable variant (target priors from confusion-matrix Black-Box Shift Estimation (40), using unlabeled target predictions only). Neither restores coverage. For three of four encoders both variants make coverage worse than doing nothing (Table IV), and for Prithvi (heaviest debt) both recover only part of the gap while remaining far below nominal and the labeled target-global arm. The overcorrection compounds at looser risk levels—at $\alpha = 0.20$ every encoder's label-shift-adjusted coverage sits $8.1$–$17.5$ pp below nominal—and more than $15$ pp below even the do-nothing source-split arm.[2] Because label-shift weighting is provably valid only when $P(x \mid y)$ is shift-invariant, this systematic failure rules out a pure class-prior-shift explanation and is consistent with a within-class conditional shift in the frozen encoders' features across regions—the geographic mechanism this paper posits, which target-side recalibration (which assumes nothing about $P(x \mid y)$) correctly repairs and label-shift reweighting cannot.

### E. Replication on a second benchmark (BigEarthNet)

We replicate the phenomenon on a second real EO benchmark: GEO-Bench (26) `m-bigearthnet` (BigEarthNet-S2 (41); 7,669 usable tiles, real lat/lon, 36.96–67.80°N). We apply two complementary analyses.

A documented dominant-label single-label proxy (12 classes, $\geq 60$ tiles each) lets the single-label LAC harness apply unchanged. All three frozen FMs replicate the pattern: ECE degrades ($0.36 \rightarrow 0.56$ Prithvi, $0.15 \rightarrow 0.42$ SSL4EO, $0.33 \rightarrow 0.59$ Clay) and spatial-Mondrian restores coverage to near-nominal at all three risk levels (Fig. 12). The proxy is a hard task (frozen-probe accuracy $0.50$–$0.61$ in-distribution, dropping to $0.27$–$0.37$ under shift), so restoration is bought at large prediction-set sizes ($\sim$6–10 of 12 classes)—a coarse, qualitative replication.

We extend the analysis with rigorous per-class binary conformal prediction on the 43-class multi-hot labels. For each of 39 well-populated labels ($\geq 120$ global positives; 27 active per seed after per-split filtering), we fit an independent binary probe (StandardScaler, L2-logistic, $C = 1$, 5 seeds) on source-region embeddings and calibrate per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ via source-only split-conformal and target-side spatial-Mondrian arms. The coverage-debt phenomenon replicates per-label: mean recall-coverage drops from 0.92 in-distribution to 0.75 under shift ($\alpha = 0.10$; mean per-label debt 0.173), with spatial-Mondrian restoring to 0.91 (per-label paired CI excludes 0 at $\alpha \in \{0.10, 0.20\}$ for all three FMs; Fig. 13). Predicted-label set sizes remain large (20.9 of 27 active labels per tile at $\alpha = 0.10$), reflecting the geographic distributional mismatch and the recall requirement. Per-label conformal is more principled than the dominant-label proxy—it offers independent per-class recall guarantees—but both yield comparably large prediction sets under this hard pan-European shift.

This conformal-risk-control (CRC) arm, which gives a finite-sample FNR guarantee, is the substrate for the paper's primary decoupling result (Table I); here we give its full per-$\alpha$ calibration and repair numbers for all five frozen encoders. Per-label split-conformal offers per-class recall calibration but no finite-sample guarantee on the example-averaged false-negative rate (FNR). We therefore add a conformal-risk-control (CRC) arm (23) on the same frozen-feature caches, splits, $4 \times 4$ Mondrian grid, and per-label probes (per-example FNR loss, monotone in the threshold $\lambda$; exact weighted-quantile calibration; 5 seeds; percentile-bootstrap CIs, $B$=2000). The three-step story repeats under the marginal-FNR guarantee for every encoder (Table VI): (i) source-calibrated CRC meets the target FNR in-distribution (e.g. $0.105$ at $\alpha$=0.10, Prithvi); (ii) the same $\widehat{\lambda}$ incurs FNR debt under the geographic shift (2–4$\times$ the nominal level, e.g. $0.206$–$0.277$ at $\alpha$=0.10); (iii) a Mondrian-CRC arm calibrated per spatial bin on the target-region slice restores marginal FNR to at-or-below nominal, at the cost of larger label sets ($\sim$17–18 vs. $\sim$8–12 labels/tile at $\alpha$=0.10). An honest efficiency note: CRC controls only the example-averaged FNR, so its in-distribution sets are smaller than per-label split-conformal's, and in-distribution FNR at $\alpha$=0.05 fluctuates slightly above nominal (mean $0.052$–$0.056$), consistent with the expectation-level $\mathbb{E}[L] \leq \alpha$ guarantee—which holds in expectation over calibration draws, not per individual run, so a single run landing marginally above $\alpha$ is expected, not a violation. Like spatial-Mondrian conformal, Mondrian-CRC requires labeled target-region calibration data—a shift-aware remedy, not a free lunch. The CRC / method comparison is summarized in Fig. 8.

[2] The label-shift run reproduces the Split and Target-global columns of Table IV within $\pm 0.001$; its own covariate-shift-weighted reference column (not shown) differs slightly because the two runs use different, both legitimate, base-rate corrections for the density-ratio discriminator. This affects only that reference column, not the label-shift arms.

TABLE IV

UNLABELED-REMEDY CONTROLS ON THE IDENTICAL EUROSAT E→W SPLIT (5-SEED MEANS; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE FROZEN RESULT RECORDS): COVARIATE-SHIFT-WEIGHTED SPLIT CONFORMAL (DENSITY-RATIO WEIGHTS FROM A SOURCE-VS-TARGET DISCRIMINATOR ON THE FROZEN EMBEDDINGS; "ESS" IS THE WEIGHTS' EFFECTIVE SAMPLE SIZE AS A FRACTION OF THE CALIBRATION SET, FIXED BEFORE CALIBRATION AND HENCE CONSTANT ACROSS $\alpha$) AND LABEL-SHIFT-WEIGHTED SPLIT CONFORMAL (ORACLE: TRUE TARGET LABEL MARGINAL; BBSE: TARGET PRIORS FROM UNLABELED TARGET PREDICTIONS). SPLIT, WEIGHTED, ESS, AND TARGET-GLOBAL COLUMNS ARE FROM THE WEIGHTED-RUN RECORD; THE LABEL-SHIFT RECORD REPRODUCES SPLIT AND TARGET-GLOBAL WITHIN $\pm 0.001$. WEIGHTED CONFORMAL UNDER-COVERS NOMINAL AT EVERY $\alpha$ FOR EVERY ENCODER (THE ESS COLLAPSE SIGNALS THE SHIFT IS NOT PURE COVARIATE SHIFT); LABEL-SHIFT ADJUSTMENT MAKES THREE OF FOUR ENCODERS *worse* THAN DOING NOTHING, AND ITS DEFICIT WIDENS AT LOOSER $\alpha$. ONLY THE LABELED TARGET-GLOBAL ARM REACHES NOMINAL.

| $\alpha$ | Encoder | Split | Weighted | ESS frac. | LS-oracle | LS-BBSE | Target-global |
|---|---|---|---|---|---|---|---|
| 0.05 | Prithvi | 0.833 | 0.919 | 1.6% | 0.879 | 0.884 | 0.952 |
| 0.05 | SSL4EO-DINO | 0.880 | 0.938 | 1.3% | 0.858 | 0.867 | 0.949 |
| 0.05 | SSL4EO-MAE | 0.899 | 0.911 | 0.4% | 0.856 | 0.865 | 0.950 |
| 0.05 | Clay | 0.918 | 0.927 | 0.1% | 0.891 | 0.895 | 0.951 |
| 0.10 | Prithvi | 0.833 | 0.881 | 1.6% | 0.793 | 0.798 | 0.899 |
| 0.10 | SSL4EO-DINO | 0.880 | 0.914 | 1.3% | 0.766 | 0.774 | 0.901 |
| 0.10 | SSL4EO-MAE | 0.899 | 0.905 | 0.4% | 0.793 | 0.803 | 0.903 |
| 0.10 | Clay | 0.918 | 0.923 | 0.1% | 0.825 | 0.832 | 0.918 |
| 0.20 | Prithvi | 0.833 | 0.855 | 1.6% | 0.675 | 0.682 | 0.833 |
| 0.20 | SSL4EO-DINO | 0.880 | 0.898 | 1.3% | 0.625 | 0.639 | 0.880 |
| 0.20 | SSL4EO-MAE | 0.899 | 0.902 | 0.4% | 0.688 | 0.704 | 0.898 |
| 0.20 | Clay | 0.918 | 0.921 | 0.1% | 0.711 | 0.719 | 0.918 |

TABLE V

BOUNDARY-SENSITIVITY SWEEP, REAL-SHIFT TARGET COVERAGE "SPLIT / TARGET-GLOBAL / SPATIAL-MONDRIAN" (5-SEED MEANS; SPLIT = SOURCE-ONLY CALIBRATION; TARGET-GLOBAL = NON-SPATIAL, TARGET-CALIBRATED ABLATION ARM; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE SAME FROZEN RESULT RECORD, EXPANDING THE $\alpha$=0.05 HEADLINE ROW). † = MONDRIAN-VS-SPLIT PAIRED CI DOES NOT EXCLUDE 0 (NO SIGNIFICANT RESTORATION). AT $\alpha$=0.05, RESTORATION HOLDS IN 15 OF 16 ENCODER×BOUNDARY CELLS (THE SOLE EXCEPTION, CLAY AT $P_{50}$, WHERE DEBT IS ≈0.014, IS MARKED); AT LOOSER $\alpha$ MOST CELLS NO LONGER NEED REPAIR BECAUSE SHIFTED ACCURACY ALREADY APPROACHES NOMINAL.

| $\alpha$ | Encoder | $P_{25}$ | $P_{33}$ (headline) | $P_{40}$ | $P_{50}$ |
|---|---|---|---|---|---|
| 0.05 | Prithvi | 0.792 / 0.946 / 0.952 | 0.833 / 0.952 / 0.955 | 0.843 / 0.950 / 0.955 | 0.859 / 0.949 / 0.952 |
| 0.05 | SSL4EO-DINO | 0.847 / 0.947 / 0.951 | 0.880 / 0.949 / 0.956 | 0.892 / 0.950 / 0.955 | 0.896 / 0.951 / 0.955 |
| 0.05 | SSL4EO-MAE | 0.865 / 0.948 / 0.957 | 0.898 / 0.950 / 0.962 | 0.910 / 0.946 / 0.962 | 0.917 / 0.952 / 0.967 |
| 0.05 | Clay | 0.897 / 0.950 / 0.955 | 0.918 / 0.951 / 0.960 | 0.926 / 0.947 / 0.960 | 0.936 / 0.950 / 0.963† |
| 0.10 | Prithvi | 0.789 / 0.899 / 0.898 | 0.833 / 0.899 / 0.911 | 0.841 / 0.902 / 0.915 | 0.854 / 0.902 / 0.917 |
| 0.10 | SSL4EO-DINO | 0.847 / 0.897 / 0.906 | 0.880 / 0.901 / 0.926† | 0.892 / 0.901 / 0.928† | 0.896 / 0.902 / 0.930† |
| 0.10 | SSL4EO-MAE | 0.865 / 0.895 / 0.920† | 0.898 / 0.903 / 0.934† | 0.910 / 0.910 / 0.938† | 0.917 / 0.917 / 0.947† |
| 0.10 | Clay | 0.897 / 0.900 / 0.920† | 0.918 / 0.918 / 0.936† | 0.926 / 0.926 / 0.935† | 0.936 / 0.936 / 0.943† |
| 0.20 | Prithvi | 0.789 / 0.802 / 0.832† | 0.833 / 0.833 / 0.856† | 0.841 / 0.841 / 0.867† | 0.854 / 0.854 / 0.877† |
| 0.20 | SSL4EO-DINO | 0.847 / 0.847 / 0.850† | 0.880 / 0.880 / 0.888† | 0.892 / 0.892 / 0.896† | 0.896 / 0.896 / 0.902† |
| 0.20 | SSL4EO-MAE | 0.865 / 0.865 / 0.880† | 0.898 / 0.898 / 0.905† | 0.910 / 0.910 / 0.917† | 0.917 / 0.917 / 0.929† |
| 0.20 | Clay | 0.897 / 0.897 / 0.897† | 0.918 / 0.918 / 0.918† | 0.926 / 0.926 / 0.926† | 0.936 / 0.936 / 0.936† |

TABLE VI
CRC arm on GEO-Bench m-bigearthnet (marginal per-example FNR; mean [95% bootstrap CI] pooled over 5 seeds). Source-calibrated CRC meets nominal in-distribution, incurs FNR debt under the real E→W shift, and target-side Mondrian-CRC restores it. Set sizes are labels/tile (of ∼27 active) under shift.

| $\alpha$ | Encoder | CRC FNR in-dist | CRC FNR shift | Mondrian-CRC FNR shift | Sets CRC/Mond. |
|---|---|---|---|---|---|
| 0.05 | Prithvi | 0.055 [0.051,0.058] | 0.130 [0.125,0.135] | 0.031 [0.029,0.034] | 15.5 / 22.3 |
| 0.05 | Clay | 0.053 [0.049,0.056] | 0.176 [0.171,0.182] | 0.022 [0.020,0.024] | 13.3 / 23.8 |
| 0.05 | SSL4EO-DINO | 0.053 [0.050,0.057] | 0.191 [0.185,0.196] | 0.032 [0.029,0.034] | 10.6 / 22.3 |
| 0.05 | SSL4EO-MAE | 0.052 [0.048,0.055] | 0.170 [0.164,0.175] | 0.030 [0.028,0.033] | 11.8 / 22.1 |
| 0.05 | DOFA | 0.056 [0.052,0.059] | 0.169 [0.164,0.175] | 0.021 [0.019,0.024] | 13.0 / 24.1 |
| 0.10 | Prithvi | 0.105 [0.101,0.110] | 0.206 [0.200,0.212] | 0.083 [0.079,0.086] | 11.9 / 18.2 |
| 0.10 | Clay | 0.108 [0.103,0.113] | 0.277 [0.271,0.283] | 0.082 [0.078,0.086] | 9.7 / 18.2 |
| 0.10 | SSL4EO-DINO | 0.107 [0.102,0.112] | 0.277 [0.271,0.283] | 0.082 [0.078,0.086] | 7.9 / 17.4 |
| 0.10 | SSL4EO-MAE | 0.108 [0.103,0.113] | 0.262 [0.256,0.268] | 0.084 [0.080,0.088] | 8.6 / 17.1 |
| 0.10 | DOFA | 0.107 [0.103,0.112] | 0.249 [0.243,0.256] | 0.080 [0.076,0.084] | 9.9 / 18.1 |
| 0.20 | Prithvi | 0.205 [0.199,0.211] | 0.346 [0.339,0.353] | 0.184 [0.179,0.190] | 8.0 / 12.9 |
| 0.20 | Clay | 0.203 [0.197,0.209] | 0.426 [0.419,0.433] | 0.185 [0.179,0.190] | 6.4 / 12.9 |
| 0.20 | SSL4EO-DINO | 0.206 [0.200,0.212] | 0.401 [0.394,0.408] | 0.191 [0.185,0.197] | 5.3 / 11.1 |
| 0.20 | SSL4EO-MAE | 0.209 [0.202,0.215] | 0.408 [0.401,0.415] | 0.189 [0.184,0.195] | 5.5 / 11.3 |
| 0.20 | DOFA | 0.203 [0.196,0.209] | 0.384 [0.377,0.391] | 0.187 [0.181,0.193] | 6.7 / 12.4 |

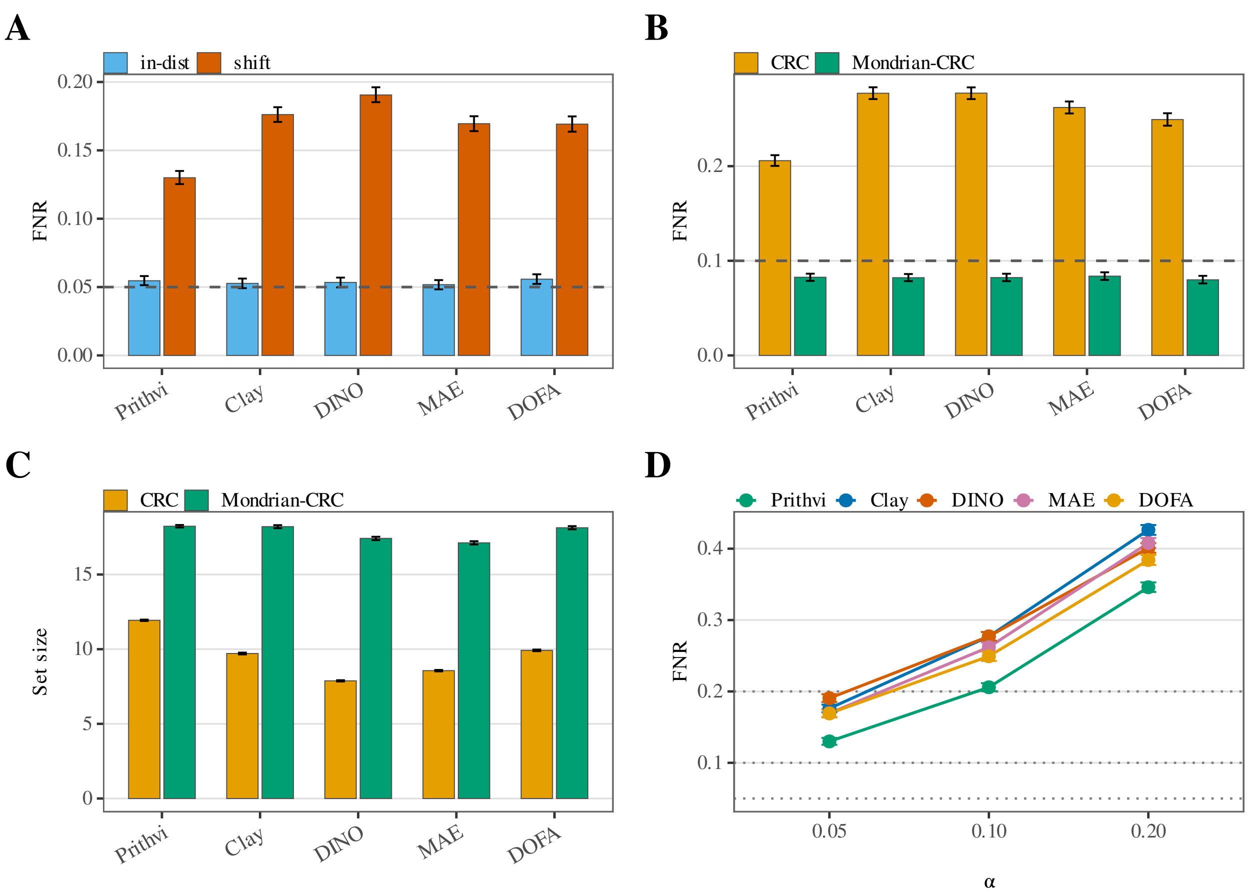


Fig. 8. CRC on BigEarthNet (frozen multi-seed bootstrap means with 95% CIs; no new runs). All panels use example-averaged FNR (or set size) pooled over five seeds; error bars are percentile bootstrap 95% intervals ($B$=2000). Encoder ticks and the (D) colour key abbreviate SSL4EO-DINO/MAE as DINO/MAE (Prithvi, Clay, DINO, MAE, DOFA). (A) Source-calibrated CRC FNR in-distribution vs. under the E→W geographic shift at $\alpha$=0.05 (blue = in-dist, vermillion = shift; dashed = nominal FNR level $\alpha$). In-distribution meets the risk level; the same $\widehat{\lambda}$ incurs large FNR debt under shift. (B) Shift FNR at $\alpha$=0.10: source-calibrated CRC vs. target-side Mondrian-CRC (orange = CRC, green = Mondrian-CRC; dashed = nominal $\alpha$). Mondrian-CRC restores FNR to at-or-below nominal for every encoder. (C) Mean set-size cost under shift at the same $\alpha$=0.10 (labels/tile of ∼27 active): Mondrian-CRC's restore comes with larger sets than source-calibrated CRC (shared orange/green method colours with (B)). (D) Source-calibrated CRC FNR under shift across $\alpha \in \{0.05, 0.10, 0.20\}$ by encoder (coloured lines; dotted horizontals = nominal $\alpha$ at each risk level).

### *F. External validation on a non-European benchmark (GEO-Bench m-so2sat)*

Both benchmarks above are European Sentinel-2. To test whether the coverage-debt phenomenon and its target-side repair survive outside Europe, we repeat the harness on GEO-Bench (26) `m-so2sat` (So2Sat LCZ42 (42); 42 cities across multiple continents; 17 local-climate-zone classes), using GEO-Bench's own train/valid (source) vs. held-out test-city (target) partition as the shift. Because this release ships no per-tile latitude/longitude in our repo, there is no spatial-Mondrian arm here; we compare only source-only split-conformal against a non-spatial, target-calibrated arm (*target-global*), i.e. exactly the label-access ablation that Table V isolated as the active ingredient on EuroSAT. All five frozen encoders now complete (Prithvi, Clay, SSL4EO ViT-S/16 MAE native-S2, SSL4EO-MAE ViT-B/16, DOFA); the earlier Clay v1.5 checkpoint-load failure ($> 4\,\mathrm{GB}$ expected in cache) was resolved on the extraction host, so the encoder-ranking diagnostic is now available on this benchmark too (all five extracted on the hash-gated frozen substrate).

The debt-and-repair pattern replicates outside Europe: under the held-out-city shift, accuracy drops by $\approx$0.14–0.18 across the five encoders ($0.670 \rightarrow 0.522$ Prithvi up to $0.796 \rightarrow 0.637$ SSL4EO-MAE)—a genuine, non-trivial distribution shift. Source-only split-conformal under-covers at every risk level and every encoder ($\alpha = 0.05$: 0.877–0.904; $\alpha = 0.20$: 0.632–0.685), and the non-spatial target-calibrated arm restores coverage to near-nominal ($\alpha = 0.05$: 0.950–0.963), with the paired-seed restoration CI excluding 0 in every one of the fifteen debt cells (5 encoders $\times$ 3 risk levels; Table VII). Restoration is bought at modest set sizes ($\sim$2–6 of 17 classes), not vacuous full-label sets. This external label-access replication now spans all five encoders on a non-European, multi-continent benchmark: it confirms the operative part of the finding—*target-side conformal recalibration*, not spatial binning, repays the debt. It still lacks a spatial-Mondrian arm (no per-tile lat/lon), so it tests the label-access mechanism only; but with all five encoders now loadable it also carries the per-class conditional-coverage diagnostic (below). A data-provenance note applies: `m-so2sat` was obtained via GEO-Bench 1.0 (HuggingFace `recursix/geo-bench-1.0`) after the designed v0.9.1 Zenodo pull was unreachable, with a label map reconstructed from per-sample ground truth (21,964/21,964 samples, 0 failures); the v0.9.1$\rightarrow$v1.0 content differences were not independently audited (see the data-provenance note in the code archive).

As a second, multiclass conditional-coverage dissociation, with all five encoders loadable, m-so2sat also lets us re-run the per-class conditional-coverage diagnostic of the EuroSAT conditional finding (§III) on a second, 17-class multiclass benchmark. At $\alpha$=0.05 we measure, for the source-only split predictor, the worst individual LCZ class coverage per encoder (Table VIII). It dissociates from both shifted accuracy and shift-ECE: $\mathrm{Spearman}(\text{worst-class cov}, \mathrm{acc}) = -0.10$ ($p$=0.87, 5 pairwise inversions) and $\mathrm{Spearman}(\text{worst-class cov}, \mathrm{ECE_{sh}}) = +0.60$ ($p$=0.29, 3 inversions); the class-coverage spread scrambles likewise ($\rho = -0.30$ against each, 6 inversions). The most accurate encoder here (SSL4EO-MAE, acc 0.637) is not the one that best protects its worst class, and the worst-covered class (0.689–0.733 across encoders, under marginal coverage 0.877–0.904) is under-served regardless of accuracy or calibration ranking. This corroborates the EuroSAT conditional-coverage finding with an independent multiclass benchmark—though on the unrepaired split sets here (no target-side repair or spatial arm on this release), and, like the EuroSAT case, it is post-hoc, exploratory, and small-$n$ ($n$=5, all $p > 0.28$; §IV-D).

TABLE VII
NON-EUROPEAN EXTERNAL VALIDATION ON GEO-BENCH m-so2sat (17 LOCAL-CLIMATE-ZONE CLASSES, 42 MULTI-CONTINENT CITIES; 5-SEED MEANS). TARGET COVERAGE "SPLIT / TARGET-GLOBAL" UNDER THE HELD-OUT-CITY SHIFT (SPLIT = SOURCE-ONLY CALIBRATION; TARGET-GLOBAL = NON-SPATIAL, TARGET-CALIBRATED LABEL-ACCESS ARM; NO SPATIAL-MONDRIAN ARM, AS THIS RELEASE CARRIES NO PER-TILE LAT/LON IN OUR REPO). SET SIZES (OF 17 CLASSES) IN BRACKETS. ALL FIVE ENCODERS NOW LOAD (CLAY V1.5 CHECKPOINT-LOAD FAILURE RESOLVED).

| $\alpha$ | Encoder | Split [set] | Target-global [set] | Acc. in→shift |
|---|---|---|---|---|
| 0.05 | Prithvi | 0.904 [3.92] | 0.956 [5.72] | 0.670→0.522 |
| 0.05 | Clay | 0.891 [3.32] | 0.954 [5.08] | 0.713→0.569 |
| 0.05 | SSL4EO ViT-S/16 MAE | 0.880 [2.42] | 0.962 [4.25] | 0.781→0.601 |
| 0.05 | SSL4EO-MAE ViT-B/16 | 0.882 [2.18] | 0.950 [3.47] | 0.796→0.637 |
| 0.05 | DOFA | 0.877 [2.49] | 0.963 [5.35] | 0.777→0.608 |
| 0.10 | Prithvi | 0.825 [2.71] | 0.907 [4.02] | 0.670→0.522 |
| 0.10 | Clay | 0.808 [2.27] | 0.888 [3.24] | 0.713→0.569 |
| 0.10 | SSL4EO ViT-S/16 MAE | 0.775 [1.67] | 0.904 [2.75] | 0.781→0.601 |
| 0.10 | SSL4EO-MAE ViT-B/16 | 0.795 [1.57] | 0.896 [2.38] | 0.796→0.637 |
| 0.10 | DOFA | 0.781 [1.69] | 0.918 [3.30] | 0.777→0.608 |
| 0.20 | Prithvi | 0.685 [1.62] | 0.819 [2.65] | 0.670→0.522 |
| 0.20 | Clay | 0.672 [1.39] | 0.795 [2.17] | 0.713→0.569 |
| 0.20 | SSL4EO ViT-S/16 MAE | 0.632 [1.08] | 0.809 [1.88] | 0.781→0.601 |
| 0.20 | SSL4EO-MAE ViT-B/16 | 0.654 [1.04] | 0.793 [1.59] | 0.796→0.637 |
| 0.20 | DOFA | 0.634 [1.07] | 0.823 [1.99] | 0.777→0.608 |

TABLE VIII
PER-CLASS CONDITIONAL COVERAGE ON GEO-BENCH m-so2sat (17 LCZ CLASSES; SOURCE-ONLY SPLIT-CONFORMAL, $\alpha$=0.05; 5 SEEDS). THE WORST INDIVIDUAL-CLASS COVERAGE DISSOCIATES FROM SHIFTED ACCURACY AND SHIFT-ECE (Spearman $= -0.10/+0.60$; $5/3$ PAIRWISE INVERSIONS), A SECOND MULTICLASS INSTANCE OF THE EUROSAT CONDITIONAL-COVERAGE COLLAPSE.

| Encoder | Acc. shift | ECE shift | Marg. cov | Worst-class cov | Class spread |
|---|---|---|---|---|---|
| Prithvi | 0.522 | 0.364 | 0.904 | 0.731 | 0.269 |
| Clay | 0.569 | 0.325 | 0.891 | 0.732 | 0.264 |
| SSL4EO ViT-S/16 MAE | 0.601 | 0.114 | 0.880 | 0.689 | 0.302 |
| SSL4EO-MAE ViT-B/16 | 0.637 | 0.173 | 0.882 | 0.702 | 0.269 |
| DOFA | 0.608 | 0.214 | 0.877 | 0.733 | 0.253 |

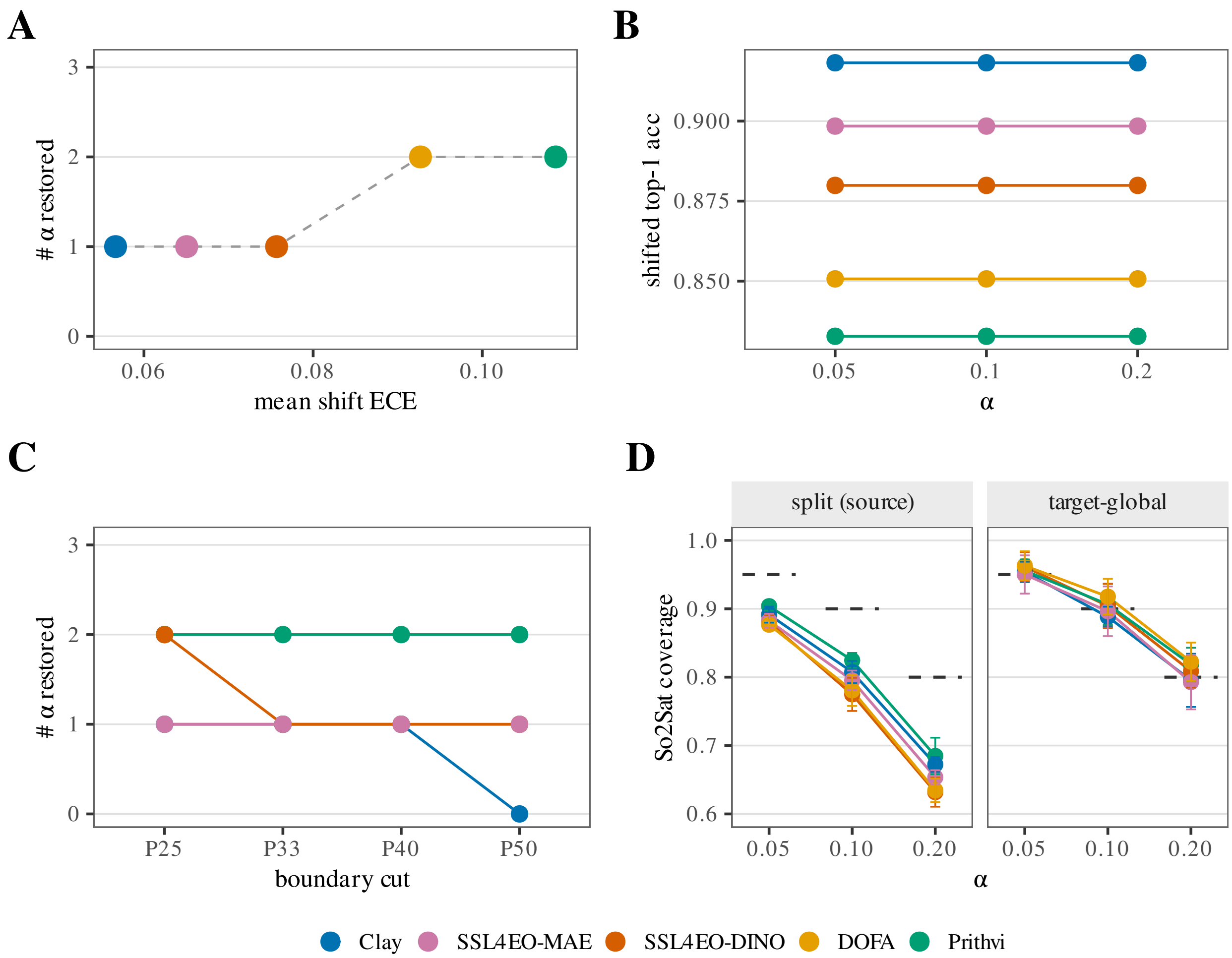


Fig. 9. Four-panel coverage-debt diagnostic across five frozen encoders (shared colour legend: Clay, SSL4EO-MAE, SSL4EO-DINO, DOFA, Prithvi). (A) Mean in-shift ECE (coverage debt) vs. the number of risk levels $\alpha \in \{0.05, 0.10, 0.20\}$ where spatial-Mondrian *restores* target coverage—i.e. the paired gap CI excludes 0 in the positive direction. Debt orders Prithvi > DOFA > SSL4EO-DINO > SSL4EO-MAE > Clay; repair counts are weakly monotone (Prithvi = DOFA 2, others 1). (B) Shifted top-1 accuracy vs. $\alpha$ (flat across risk levels; ranking matches debt). (C) Boundary-sweep check on the four battery encoders (DOFA absent from that frozen sweep): #$\alpha$ restored at spatial cuts P25–P50. (D) So2Sat LCZ cross-continent label-access check: source-split vs. target-global coverage (facets); dashed horizontal guides mark nominal $1-\alpha$ at each $\alpha$ (not error bars); vertical whiskers are seed-level CIs. Set-size cost of the same repair appears in the companion four-panel set-size figure.

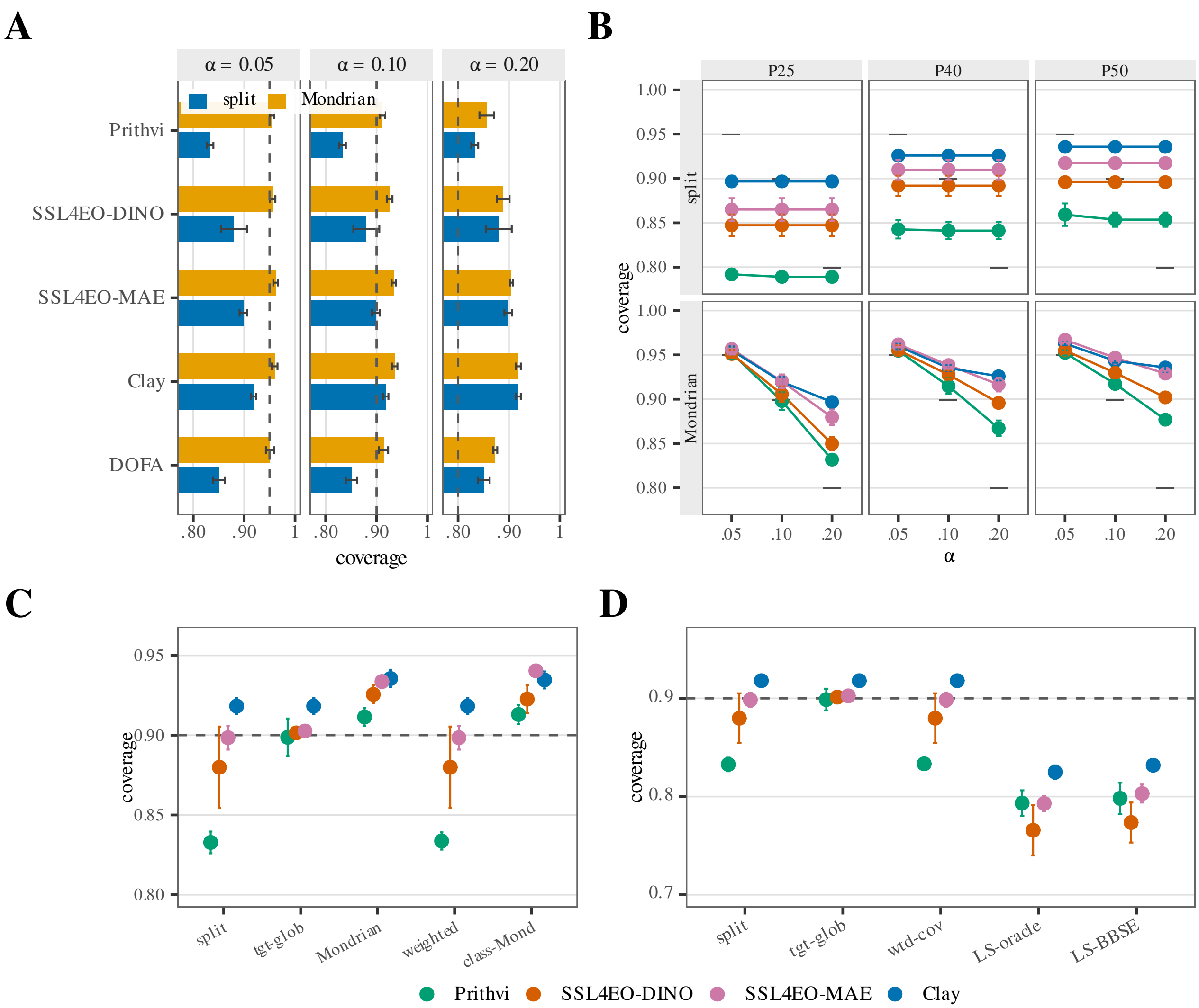


Fig. 10. Coverage restoration under real geographic shift (error bars = seed-level $t$-CIs, $n$=5; dashed = nominal $1-\alpha$). (a) EuroSAT $P_{33}$: source split-conformal vs. spatial-Mondrian coverage by encoder and $\alpha$ (five encoders, including DOFA). (b) Same split vs. Mondrian contrast at non-$P_{33}$ spatial cuts (P25/P40/P50); row = arm, colour = encoder (four-encoder battery; DOFA absent from this frozen record). (c) Five conformal arms at $\alpha$=0.10 (ticks: split, tgt-glob, Mondrian, weighted, class-Mond). (d) Label-shift controls at $\alpha$=0.10 (ticks: split, tgt-glob, wtd-cov, LS-oracle, LS-BBSE). Spatial-Mondrian restores where debt exists; DOFA's shifted split coverage is 0.851 at every $\alpha$ and spatial-Mondrian lifts it to 0.951/0.913/0.873.

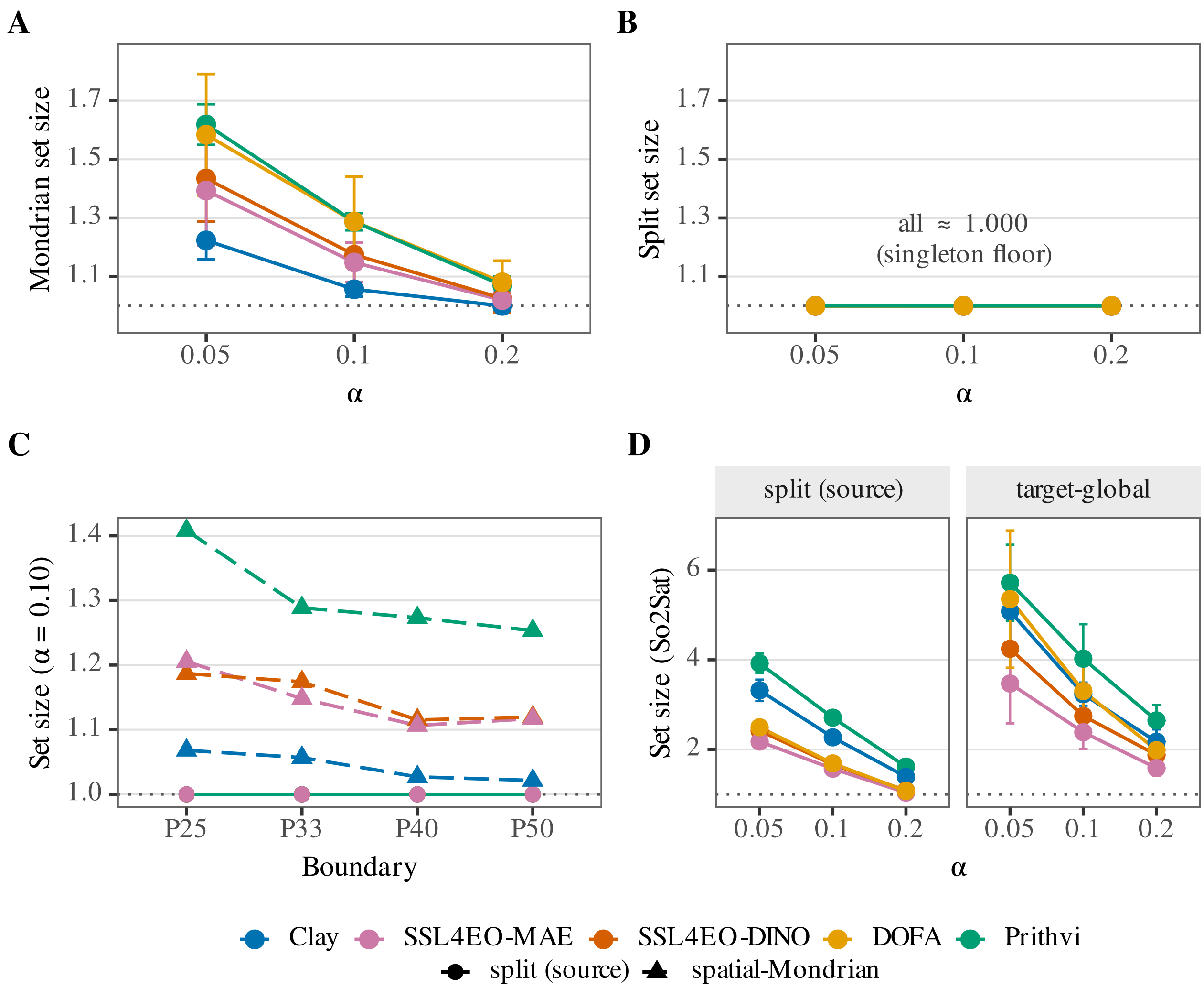


Fig. 11. Four-panel set-size cost of restoration (error bars seed-level $t$-CIs, $n$=5; dotted line = singleton floor of set size 1). Shared encoder colours throughout; panel (c) uses linetype/shape for split vs. spatial-Mondrian. (a) EuroSAT spatial-Mondrian mean set size vs. $\alpha$ (rises above the floor only where Mondrian repays coverage debt; DOFA 1.58/1.29/1.08 at $\alpha$ = 0.05/0.10/0.20). (b) Source split-conformal set size on the *same* $y$-scale as (a): every encoder sits on the singleton floor (≈ 1.000), so the excess size in (a) is the restoration cost. (c) Set size vs. spatial boundary percentile at $\alpha$=0.10 (solid = source split; dashed = spatial-Mondrian). (d) So2Sat LCZ (17 classes) source-split vs. target-global set size in side-by-side facets—larger absolute sets than EuroSAT, on its own $y$-scale.

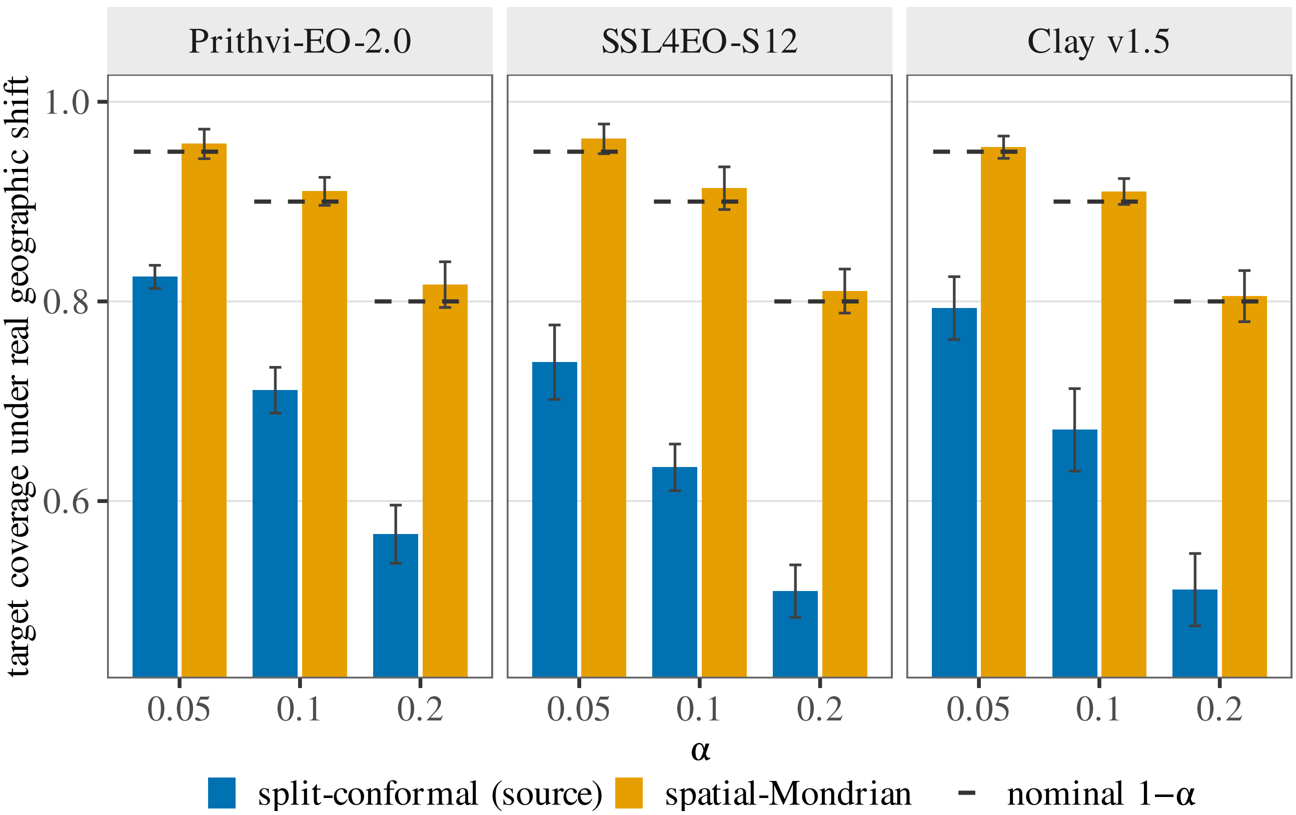


Fig. 12. Dominant-label proxy on GEO-Bench `m-bigearthnet`: multi-hot BigEarthNet-S2 tiles reduced to the most frequent of 12 well-populated classes under a real geographic E→W shift (single-label LAC harness unchanged). Columns are the three frozen encoders (Prithvi-EO-2.0, SSL4EO-S12, Clay v1.5). Within each panel, blue bars are source-only split-conformal and orange bars are target-side spatial-Mondrian at $\alpha \in \{0.05, 0.10, 0.20\}$. Short black dashed segments mark the nominal coverage level $1-\alpha$ at each $\alpha$ (legend: "nominal $1-\alpha$"); vertical whiskers on the bars are seed-level Student $t$ intervals ($n$=5), not the same mark. Coverage debt and Mondrian restoration replicate across all three encoders. This proxy is a coarse, qualitative replication; the companion per-label multi-label analysis is the primary BigEarthNet evidence.

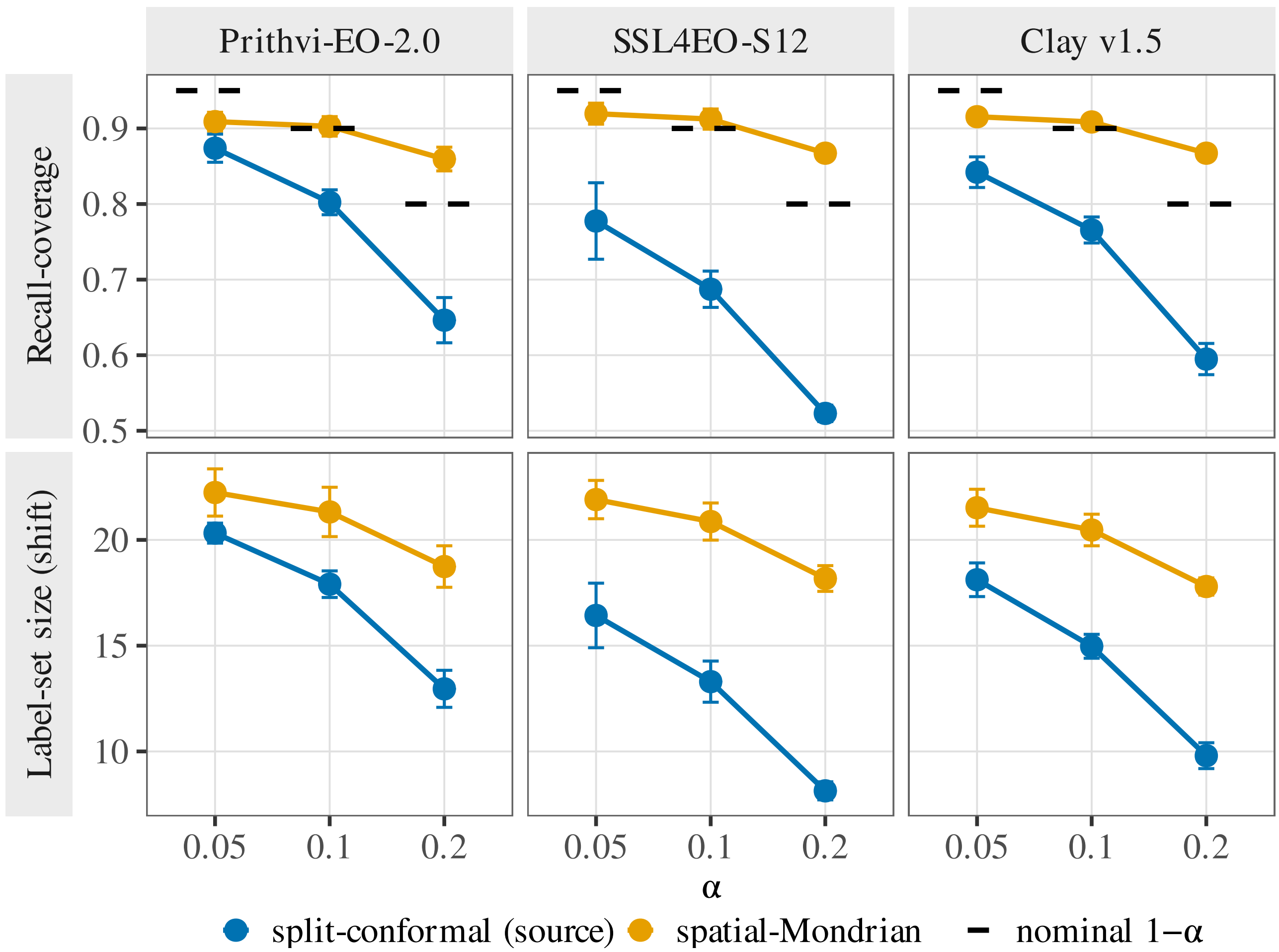


Fig. 13. Per-label multi-label conformal on GEO-Bench `m-bigearthnet` (BigEarthNet-S2; 27 active labels after per-split filtering; real geographic E→W shift). Columns are frozen encoders; rows are (top) mean per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ and (bottom) mean predicted label-set size per tile. Blue = source-only split-conformal; orange = target-side spatial-Mondrian; $\alpha \in \{0.05, 0.10, 0.20\}$. Short black dashed segments in the top row are nominal $1-\alpha$ guides (legend: "nominal $1-\alpha$"), not error bars; vertical whiskers on the points are seed-level Student $t$ intervals ($n$=5). Mondrian restores per-label coverage at $\alpha \in \{0.10, 0.20\}$ (paired CI across labels excludes 0) for all three FMs. Relative to the dominant-label proxy, this per-label analysis is primary: it supplies independent per-class recall guarantees rather than a single-label reduction.

## IV. Discussion

### A. Interpretation

A conformal audit of a frozen GFM earns its keep where accuracy and calibration go silent (37, 43). On multi-label FNR risk control, every one of thirteen frozen encoders incurs a geographic FNR debt under shift (2.6–$4.7\times$ nominal) that only a labeled target slice repairs; accuracy does not reliably identify the most robust encoder—the least accurate is the most robust, yet at $n$=13 the rank correlation is only a weak positive ($\rho = +0.32$; Table I). On per-class conditional coverage, restoring the marginal guarantee can still leave one land-cover class covered at $0.850$ under a $0.956$ marginal (Table II), again unpredictable from the scalars a practitioner would rank on. Symmetrically, the audit adds little in the high-accuracy singleton regime: marginal set size is then a deterministic restatement of accuracy, so EuroSAT at full data is a mapped negative control rather than a decoupling, and the preregistered efficiency-decoupling route is falsified. Within that boundary, the descriptive coverage-debt ranking remains useful—ordering Prithvi, SSL4EO-DINO, SSL4EO-MAE, and Clay consistently across risk levels—but we avoid a “universal fix” claim: where shifted accuracy already meets the nominal level, there is no debt and no repair. Relative to accuracy-centric GFM surveys and benchmarks (44, 26, 45), the increment is pricing geographic coverage debt rather than OOD accuracy alone.

### B. Mechanisms and controls

Headline coverage and ranking numbers use an unadjusted logistic probe and therefore inherit the $P_{33}$ source/target class-prior mismatch (Methods; up to $6.54\times$ enrichment / $\sim 4.7\times$ depletion for individual EuroSAT classes; Fig. 14).

TABLE IX
EUROSAT-MS PER-CLASS FREQUENCY AT THE $P_{33}$ LONGITUDE CUT (SOURCE = E/CENTRAL EUROPE, $n$=18,090; TARGET = W/IBERIAN-ATLANTIC EUROPE, $n$=8,910). CLASSES NOT LISTED SHIFT BY <2.4×.

| Class | Source % | Target % | Target/Source ratio |
|---|---|---|---|
| 5 (Pasture) | 2.62 | 17.13 | 6.54× (enriched) |
| 2 (HerbaceousVegetation) | 7.68 | 18.07 | 2.35× (enriched) |
| 9 (SeaLake) | 14.10 | 5.04 | 0.36× (depleted) |
| 8 (River) | 11.60 | 4.51 | 0.39× (depleted) |
| 1 (Forest) | 15.02 | 3.16 | 0.21× (∼4.7× depleted) |

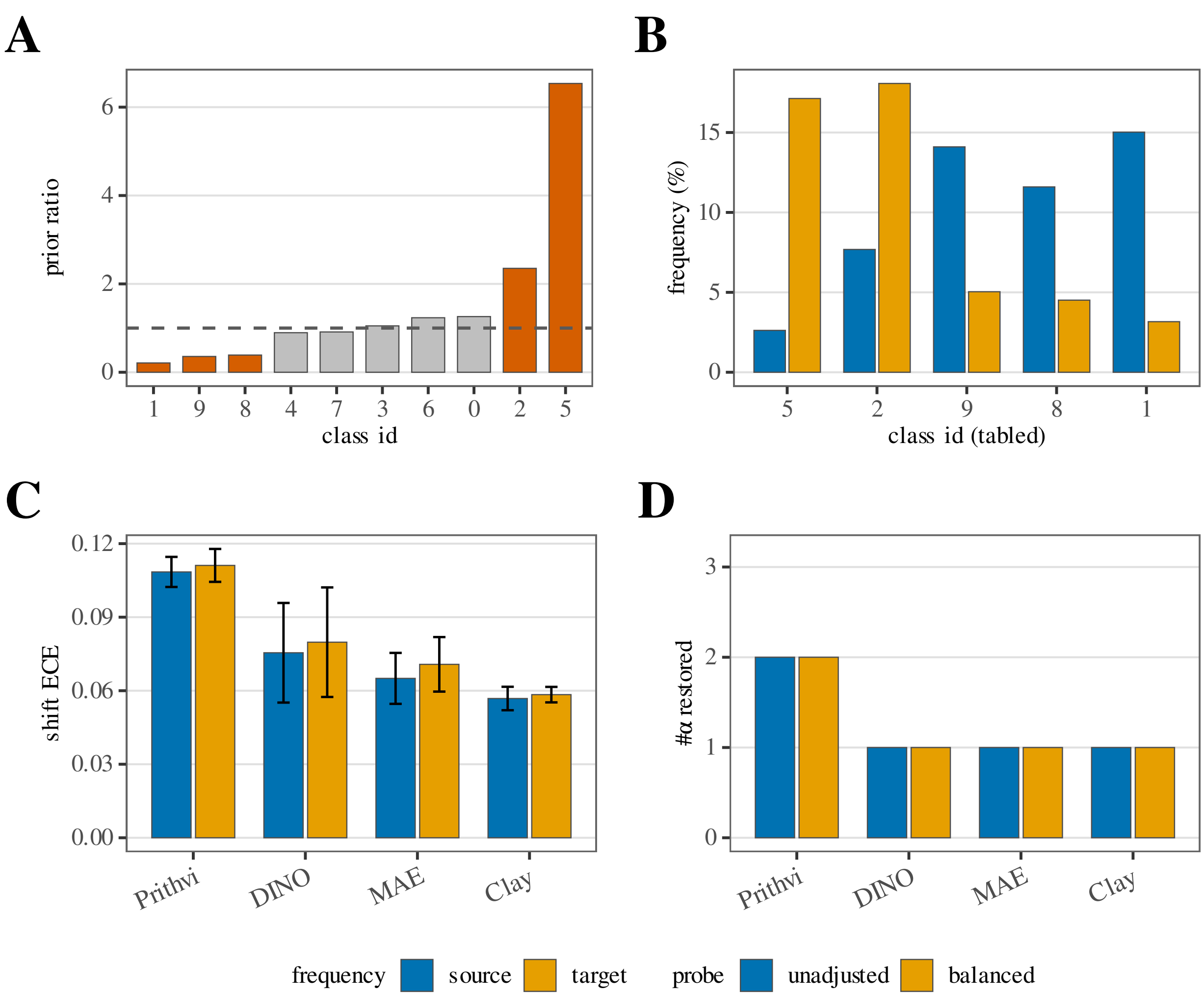


Fig. 14. Class-prior / balanced-probe control at the frozen $P_{33}$ cut (no new runs). **A:** target/source class-prior ratio by EuroSAT class id (orange = five extreme classes; grey = others); dashed line at ratio 1 (no prior shift). **B:** source vs. target class frequency (%) for those five classes. **C:** shift ECE at $\alpha$=0.05 under the unadjusted (headline) vs. class-balanced probe; error bars are 5-seed $t$-CIs. **D:** count of $\alpha \in \{0.05, 0.10, 0.20\}$ at which spatial-Mondrian restores coverage (paired restoration CI excludes 0)—the "#$\alpha$ repaired" count, not Mondrian coverage itself. Shared bottom legend: frequency (B) = source/target; probe (C–D) = unadjusted/balanced (same blue/orange hues, distinct meanings). Principle: the balanced probe is a *control*—encoder ranking by shift-ECE and repair counts should be stable if the $P_{33}$ class-prior mismatch is not driving the finding.

Refitting every probe with class-balanced sample weighting on the identical cached embeddings leaves the shift-ECE ranking, the per-encoder repair count, and the analogous per-label-balanced BigEarthNet check unchanged to within seed noise (Table X). The prior mismatch therefore does not appear to drive the ranking; the balanced control was run only at the frozen $P_{33}$ boundary (Table V remains unadjusted). The ViT-B backbone (SSL4EO-MAE) incurs less debt than the ViT-S backbone (SSL4EO-DINO), but those checkpoints also differ in objective and release, so the pairwise ordering is consistent with—yet does not isolate—a backbone-capacity mechanism. EO conformal applications establish distribution-free sets for EO pipelines (13, 14, 15) but do not quantify region-conditional debt for interchangeable frozen GFMs; we reuse split and spatial-Mondrian conformal prediction with CRC (21, 33, 23) under a frozen-probe constraint rather than proposing a new estimator.

### C. Deployment implications

Operationally, the report card is a pre-deployment gate. If the task is multi-label (or otherwise below the singleton-degeneracy boundary), treat in-distribution CRC/Mondrian coverage as necessary but not sufficient: budget a labeled target-region calibration slice before claiming a reliability guarantee, because unlabeled covariate-shift weighting collapses on effective sample size here and label-shift reweighting does not substitute (16). Prefer encoder selection that jointly inspects set-size / FNR robustness and worst-class conditional coverage, not accuracy or ECE alone (34, 46). Table XI illustrates the gate for the median-accuracy encoder (Clay v1.5; mAP $0.502$, rank 3 of 5) at $\alpha$=0.05: uncalibrated example-averaged FNR misses the $0.05$ target by $\approx 3.5\times$ (0.176); a $30\%$ labeled target slice with Mondrian CRC restores FNR to $0.022$ at the cost of larger sets ($13.3 \to 23.8$ of 27 labels). On `m-so2sat` (LCZ-17), a healthy marginal still hides a worst class at $0.732$. Deploy only with labeled target recalibration, and monitor per-class coverage rather than the marginal alone. Relative to geographic-robustness testbeds that score OOD accuracy rather than coverage debt (47, 48, 49), the practical increment is explicit pricing of that labeled target slice.

### D. Scope and limitations

The single preregistered confirmatory result is a falsification: Stage 2 required spatial-Mondrian restoration at $\geq 2$ of 3 risk levels for both Prithvi and Clay; Clay restored at only $1/3$ (Prithvi $2/3$). The encoder-dependent, debt-targeted account used throughout is therefore an exploratory reframe; we introduce no new estimator, only the report-card protocol (§IV-C) and the audited evidence behind it (21, 23). Beside that negative, a post-hoc sign test over the thirteen frozen per-encoder FNR estimates confirms the within-encoder debt-and-repair pattern at $p \approx 2.4 \times 10^{-4}$ in both directions (§III).

The three re-centered decoupling axes are post-hoc and, for the conditional analyses, small-$n$. Roster expansion weakened the cross-encoder FNR claim: five-encoder $\rho = +0.90$ (exact permutation $p$=0.083) regresses to $+0.32$ at $n$=13 (permutation $p$=0.28; $48/78$ pairwise inversions). We keep only the directional claim that accuracy is not a reliable guide—the extreme and the near-calibration-matched Prithvi/Clay pair still dissociate—plus the surviving FNR-vs-set-size tradeoff ($\rho = -0.66$, $p$=0.017 at $\alpha$=0.10). Conditional-coverage dissociation ($n$=5, all $p > 0.28$) supports "no detectable dependence," not proven independence.

Scope is optical Sentinel-2 with a EuroSAT-MS longitude cut, BigEarthNet replication, and one non-European LCZ extension (`m-so2sat`)—not continent-scale spatial-Mondrian generalization, not SAR/optical cross-sensor transfer (a preliminary S1 probe failed in-distribution control and is withheld), and not a 2025-generation leaderboard (8, 7). Covariate-shift-weighted and label-shift conformal controls fail to restore coverage (Tables IV, IV); class-conditional Mondrian remains unexecuted. The planning curve (Fig. 7) and boundary sweep (Table V) are post-hoc extensions of the frozen $30\%$ / $P_{33}$ protocol. Honest follow-ons include expanding the frozen roster and geographic axes, testing whether audited or adaptive remedies (50, 51, 52) reduce the labeled target budget, and pairing the report card with OOD flags and area-of-applicability maps (10, 49, 11).

## Data and Code Availability

EuroSAT-MS (public, Sentinel-2); frozen weights Prithvi-EO-2.0-300M, Clay v1.5, SSL4EO-S12 DINO (torchgeo), SSL4EO-S12 MAE (`wangyi111/SSL4EO-S12`, HuggingFace), all public/no-auth. The code and frozen result records used to produce every result and figure in this manuscript are available at GitHub (https://github.com/PeterPonyu/geospatial-fm-reliability-research; repository handle `PeterPonyu`) and archived at Zenodo, DOI `10.5281/zenodo.21130299` (this DOI is currently reserved on a draft deposition and becomes active when the record is published); reproduction instructions and a fast regression gate are provided in the code archive. Per IEEE's author-reproducibility guidelines, code archiving with a persistent identifier via GitHub plus a DOI-issuing repository (Zenodo) is sufficient for TGRS; IEEE's optional Code Ocean partnership was not used for this submission.

TABLE X

CLASS-PRIOR-SHIFT CONTROL: CLASS-BALANCED SAMPLE WEIGHTING VS. THE PAPER'S UNADJUSTED LOGISTIC PROBE, RE-FIT ON THE IDENTICAL CACHED FROZEN EMBEDDINGS (NO GPU RE-EXTRACTION), 5 SEEDS, REAL E→W SHIFT. TOP PANEL (EUROSAT, $\alpha = 0.05$): ECE-SHIFT RANKING AND PER-ENCODER REPAIR COUNT (OF 3 $\alpha$'S) ARE UNCHANGED UNDER THE BALANCED CONTROL. BOTTOM PANEL (BIGEARTHNET-S2, $\alpha = 0.10$, PER-LABEL CONFORMAL): THE ANALOGOUS BALANCED-PROBE REFIT OF THE PER-LABEL HARNESS (PREVIOUSLY REPORTED ONLY IN PROSE) LEAVES MEAN PER-LABEL DEBT AND MONDRIAN RESTORATION ESSENTIALLY UNCHANGED FOR ALL THREE AVAILABLE FMS (SSL4EO-MAE WAS NOT RUN IN THE PER-LABEL BIGEARTHNET HARNESS).

| | Unadjusted (paper headline) | | | Balanced control | | |
|---|---|---|---|---|---|---|
| Encoder | Acc. shift | ECE shift | #$\alpha$ repaired | Acc. shift | ECE shift | #$\alpha$ repaired |
| Prithvi | 0.833 | 0.1085 | 2/3 | 0.835 | 0.1111 | 2/3 |
| SSL4EO-DINO | 0.880 | 0.0755 | 1/3 | 0.878 | 0.0798 | 1/3 |
| SSL4EO-MAE | 0.898 | 0.0650 | 1/3 | 0.895 | 0.0708 | 1/3 |
| Clay | 0.918 | 0.0568 | 1/3 | 0.917 | 0.0584 | 1/3 |

| | Unadjusted (paper headline) | | Balanced control | |
|---|---|---|---|---|
| Encoder | Mean per-label debt | Mondrian restores | Mean per-label debt | Mondrian restores |
| Prithvi | 0.117 | Yes | 0.113 | Yes |
| SSL4EO-DINO | 0.239 | Yes | 0.236 | Yes |
| Clay | 0.163 | Yes | 0.161 | Yes |

TABLE XI

DEPLOYMENT RELIABILITY REPORT CARD (ILLUSTRATION; POST-HOC) FOR A MEDIAN-ACCURACY FROZEN ENCODER (CLAY V1.5) MONITORING LAND COVER IN AN UNSEEN REGION AT $\alpha$=0.05. MULTI-LABEL FNR ROWS: GEO-BENCH `m-bigearthnet`, $E \to W$ SHIFT; WORST-CLASS ROW: GEO-BENCH `m-so2sat` LCZ-17 (SPLIT ARM). BRACKETS ARE SEED-LEVEL $t$-CIS ($n$=5). ALL VALUES TRACE TO THE FROZEN RESULT RECORDS DESCRIBED IN THE CODE ARCHIVE.

| Report-card question ($\alpha$=0.05, encoder Clay) | Value |
|---|---|
| Multi-label FNR if deployed uncalibrated | $0.175$ $[0.143, 0.208]$ |
| (relative to the 0.05 target) | $\approx 3.5\times$ over |
| FNR after a 30% target slice (Mondrian CRC) | $0.022$ $[0.015, 0.029]$ |
| Efficiency cost (mean set size, of 27 labels) | $13.3 \to 23.8$ |
| Worst class, 2nd region (`m-so2sat` LCZ) | $0.732$ $[0.692, 0.772]$ |
| (under a marginal coverage of) | 0.891 |

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